



CH450

Bioengineering

Modelling & Analysis



CH450 BIOENGINEERING MODELLING & ANALYSIS

ABET eCOURSEBOOK

AY2021-2026

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UNITED STATES MILITARY ACADEMY
WEST POINT.

SECTION I

COURSE ASSESSMENTS



TAB A AY2026 COURSE ASSESSMENT

Department of Chemical and Biological Science and Engineering
United States Military Academy
West Point, New York 10996

MADN-CBSE

26 May 2026

MEMORANDUM FOR Chemical Engineering Program Director, Department of Chemical and Biological Science and Engineering, United States Military Academy, West Point, New York 10996

SUBJECT: CH450 Bioengineering Modeling, Analysis & Design Course Assessment (AY26-2)

1. The course assessment meeting for CH450 was conducted on 26 May 2026. Attendees: Dr. Simuck F. Yuk, Dr. Enoch A. Nagelli, Dr. Andrew I. Biaglow, and COL Corey M. James. The topics covered included: 1) expansion of the in-class coding exercises with the contemporary subject matters; 2) consideration of oral examination in CH450. Summary of recommended changes for AY27 is included in Section III.
2. The CH450 course assessment in accordance with the course director's handbook from AY26 is attached.
3. POC is the undersigned.

CF: Dr. Biaglow, Dr. Nagelli, COL James

Dr. Simuck F. Yuk
Associate Professor, Chemical Engineering
CH450 Course Director

SECTION I: COURSE DESCRIPTION

1. Current Red Book Description.

CH450 (Version: 2019 2) COURSE DETAILS

COURSE	TITLE	EFF YEAR	EFF TERM	CREDIT HOURS
CH450	BIOENGINEERING MDLING & ANLSIS	2019	2	3.0 (BS=0.0, ET=3.0, MA=0.0)

SCOPE

This course provides a broad understanding of bioengineering disciplines to include biomechanics, biomaterials, tissue engineering, biocatalysis, biochemical engineering, and biosensors. Fundamental concepts of molecular kinetics, thermodynamics, and mass transport are applied in problem sets in each bioengineering sub-discipline and capstone design project providing students the opportunity for modeling, analysis, and design from the biomolecular to physiological length scale and across multiple time scales. Modeling software such as MATLAB and Mathematica are extensively used.

LESSONS: 30 @ 75 min (2.000 Att/wk) **LABS:** 0 @ 0 min

SPECIAL REQUIREMENTS:

None

CH350 COURSE REQUISITES

TYPE	COURSE	EFF YEAR	EFF TERM	TRACK	RED BOOK FLG
PRE-REQUISITE	CH102	2019	2	10	Y
	MA205	2017	1	9	Y
	MA255	2016	2	10	Y
	PH202	2023	1	8	Y
	PH252	2023	1	1	Y

2. Enrollment Numbers.

- AY 2025-2: 16
- AY 2026-2: 18
- AY 2027-2: 18 (projected)
- AY 2028-2: 7 (projected)

3. Current Course Objectives.

This course provides a broad understanding of bioengineering disciplines to include biomechanics, biomaterials, tissue engineering, biocatalysis, biochemical engineering, and biosensors. Specifically, the objectives of the course are:

- 1) Understand the broad meaning of the term "bioengineering" and the interface between research, engineering, and clinical fields.
- 2) Synthesize math, science, and engineering concepts from the Core Sequence and major courses in a biological/bioengineering context.
- 3) Understand, apply, and manipulate models for bioengineering design.
- 4) Reinforce and strengthen conceptual and practical understanding of fundamental thermodynamics, kinetics, and mass transport.
- 5) Further develop critical thinking skills using journal articles.
- 6) Develop oral and written communication skills and continue to develop cadets' ability to prepare technical reports.

4. Survey Questions. Below are questions to the web-based survey of questions asked to CH450 cadets at the end of the AY26-2 semester:

Academy

Q1. My fellow students contributed to my learning in this course.

Q2. My motivation to learn and to continue learning has increased because of this course.

Q3. My personal schedule allows me enough time to reflect on the material I have learned in class.

Q4. My personal schedule allows me enough time to adequately prepare for my optimum academic performance.

Academy - I

Q5. This instructor encouraged students to be responsible for their own learning.

Q6. This instructor used effective techniques for learning, both in class and for out-of-class assignments.

Q7. My instructor cared about my learning in this course.

Q8. My instructor demonstrated respect for cadets as individuals.

Cognitive Domain

Q9. In this course, my critical thinking ability increased.

Q10. The homework assignments, papers, and projects in this course could be completed within the USMA time guideline of a 2:1 ratio of out-of-class time versus in-class time.

Q11. The classroom environment (e.g., desk setup, boards, technology, lights, etc.) had a positive impact on my ability to learn.

Q12. In this course, my ability to think creatively and take intellectual risks increased.

Q13. This instructor stimulated my thinking.

Q14. This instructor encouraged students to think creatively and take intellectual risks.

Department

Q15. Consider the departments responsible for the courses which you have taken this semester. Relative to other departments, my experience with the Department of Chemical and Biological Science and Engineering was very satisfying.

Q16. The laboratory program was useful in supporting, reinforcing, or augmenting my understanding of course content.

Q17. The course text and references were very helpful in my gaining understanding of the course material.

Q18. The Course Goals and Lesson Objectives for this course gave me a clear understanding of what was expected of me to do well in this course.

Q19. The WPRs, block exams, and quizzes in this course were a reasonable evaluation of my understanding of the course material.

5. Course GPA. The GPA for the course from the last 3 years follows below.

AY26-2 (Spring 2026)	3.74
AY25-2 (Spring 2025)	3.55
AY24-2 (Spring 2024)	3.61

6. TEE Grade.

No TEE implemented in CH450 for AY26-2.

7.a. Current Textbook.

Enderle, John, and Joseph Bronzino. Introduction to biomedical engineering. Academic press, 2011, ISBN: 978-0-12-374979-6.

7.b. Course Content.

Part I: Biomechanics

Part II: Biomaterials

Part III: Tissue Engineering

Part IV: Biochemical Reactions and Enzyme Kinetics

Part V: Biosensors and Bioimaging

7.c. Lesson Sequence

LESSON	TOPICS
Part I Biomechanics	
1	Introduction & Modeling
2	Biomechanics I
3	Biomechanics II
4	Biomechanics III
5	Modeling: Biomechanics
Part II Biomaterials	
6	Biomaterials I
7	Biomaterials II
8	Biomaterials III
9	Biomaterials IV
10	Modeling: Biomaterials
Part III Tissue Engineering	
11	Tissue Engineering I
12	Tissue Engineering II
13	Tissue Engineering III
14	Tissue Engineering IV
15	Modeling: Tissue Engineering
Part IV Biochemical Reactions and Enzyme Kinetics	
16	Compartment Modeling
17	Biochemical Enzyme Kinetics I
18	Biochemical Enzyme Kinetics II
19	Biochemical Enzyme Kinetics III
20	Biochemical Enzyme Kinetics IV
21	Modeling: Biochemical Enzyme Kinetics
Part V Biosensors and Bioimaging	
22	Biomedical Sensors I
23	Biomedical Sensors II
24	Special Topic: Guest Lecture
25	Bio-Imaging I
26	Bio-Imaging II
27	Modeling: Biomedical Sensors
28	Course Review

29	Capstone Presentations I
30	Capstone Presentations II

7.d. Summary of Graded Requirements

Requirement	Number	Points	Total	%
Problem Sets	5	100	500	50%
Quizzes	5	30	150	15%
Capstone Presentation/IPR	1	100	100	10%
Capstone Paper	1	150	150	15%
Instructor Points	---	100	100	10%
TOTAL			1000	100%

7.e. Areas of Special Emphasis.

Course Director's Area of Special Emphasis

My primary area of interest was Bioengineering modeling and analysis. This course emphasized the development of mathematical models and how these models can be applied to analyze the quantitative results from various biological systems. A large portion of knowledge was geared toward translating biological concepts into mathematical forms and creating working models. I wanted cadets to understand how to develop reasonable governing equations by assuming the appropriate initial and boundary conditions, so they can obtain and analyze the quantitative results as an engineer.

My second area of special emphasis was on the design of a fermenter utilizing biological components. The problems demonstrated by the instructor in class centered around designing a reactor system that integrates material balance across cell, substrate, and production concentrations. Throughout the course, various bioreactor designs were introduced, highlighting the distinct reactor conditions present in batch and continuous cultivation.

My third area of special emphasis was on the ability to communicate effectively. As part of the capstone project, cadets are required to individually submit a technical abstract and subsequently produce a final research paper. Part of the grade on professionalism of the written report is devoted to their communication skills. Cadets are evaluated on their technical writing and oral presentation skills, along with the feedback on different aspects of capstone projects.

My fourth area of special emphasis was on the use of modern computing techniques. Some of the kinetic problems in this course involved higher level calculus that required the use of an ODE solver, i.e., Mathematica, to project the kinetic trends in efficient manners. Several problems, i.e., solving Michaelis Menten kinetics for different species presented in the chemostat, were introduced with the emphasis of using computing codes to approach numerical solutions.

8. Contribution to Student Outcomes

The following list contains the Chemical Engineering Student Outcomes. The phrases in **bold-faced red font** are the topics that this course is specifically asked to accomplish by the program. These bullets are also indicated with **red dashed boxes** in Section II.8

The Chemical Engineering Program provides the graduate with a thorough grounding and working knowledge of ...

- General, organic, and physical chemistry.
- **Material and energy balances as well as safety and environmental factors.**
- Thermodynamics of physical and chemical equilibria.
- Heat, mass, and momentum transfer.
- **Chemical reaction engineering.**
- Continuous and staged separation operations.
- Process dynamics and control.
- Modern experimental and computing techniques.
- Process design.

9. Resources and Laboratories.

9a. Laboratories. There were no wet labs conducted on this course.

9b. Computer Labs. All class lectures meet at BH630. No labs were executed in CH450.

9c. Physical Models & Demos. There are no physical models or demos used in the course.

9d. Technician Support. IT support from the CBSE IT office for maintenance of computers in BH630 was timely and addressed any issues that arose. In future considerations for distance learning, a scanner, Epson elmo, and a second monitor are important to have at designated location of teaching.

9e. Supplies. Mathematica 11 and Microsoft Office (Word, PowerPoint, Excel, and Explorer).

9f. Additional Facilities. None.

9g. Unfunded Requests. None.

10. Recommendations from AY24-2 and Implementation in AY25-2:

<i>Section I.</i>	<i>Recommendation (AY24-2)</i>	<i>Implementation (AY25-2)</i>
1. Redbook	No change.	Implemented as recommended (IAR)
2. Enrollment	No change.	No change.
3. Course Objectives	No change.	No change.
3.A. Coverage	No change.	No change.
3.B. Performance	No change.	No change.
4. Survey Questions	No change.	The end-of-course survey, administered through CoursEval, achieved a 76%

		response rate, marking a significant increase from the 40% recorded in AY25-2. This substantial growth is primarily attributed to the introduction of extra instructor points (20 pts) as an incentive
5. Course QPA	No change.	No change.
6. TEE Grades	No change.	No change.
7. Course Processes 7.A. Textbook	No change.	No change.
7.B. Course Content	No change.	No change.
7.C. Lessons and Labs	No change.	No change.
7.D. Graded Events	No change.	A new type of problem was integrated into in-class activities to better increase cadets' understanding of the learning objectives. Aligning these exercises with contemporary, real-world topics enhanced classroom engagement and captured student interest.
7.E. Areas of Special Emphasis	No change.	IAR
8. Contribution to Student Outcomes 8.A. Coverage	No change.	IAR
8.B. Performance	No change.	IAR
9. Resources and Laboratories 9.A. Laboratories	No change.	IAR
9.B. Computer Labs	No change.	IAR
9.C. Physical Models & Demos	No change.	IAR
9.D. Technician Support	No change.	IAR
9.E. Supplies	No change.	IAR

9.F. Additional Facilities	No change.	IAR
9.G. Unfunded Requests	No change.	IAR

SECTION II - COURSE ASSESSMENT

1. Red Book Description. The Red Book description was followed verbatim.

2. Enrollment. The enrollment on the course was 18 for AY26-2, with the projected enrollment on AY27-2 to be 18. During Term 26-2, no cadets failed the course, and none were at serious risk of failing the course going into the Term End.

3. Course Objectives. The course objectives were covered thoroughly and achieved with proficiency. The course objectives cover the body of knowledge appropriately. The course objectives lend themselves to assessment.

3.b. Coverage

Coverage of course objectives is presented in Figure 1. As in the other courses in this program, coverage is assessed by the course director, who assigns a 1 or a 0 to a graded event or portion of a graded event. A score of 1 on a graded event or a portion of a graded event designates significant coverage, while a 0 designates an insignificant amount or no coverage. Coverage scores are then averaged across all similar graded events in the course (problem sets, exams, etc.). A score close to 1 indicates a relatively large portion of that type of graded event covered that objective. For example, a score of 0.6 on design problems under Objective 1 (mass/energy balances) indicates that 60% or 3 out of 5 of the problem sets contained coverage of that objective. This does not say that the other problem set had zero coverage of that objective, only that it was not counted by the course director for assessment. An objective with multiple bars shows that multiple types of graded events covered that objective. Raw data are included in the spreadsheet attached to the end of this document in Attachment 3. Because of this analysis, we conclude that course objectives were covered.

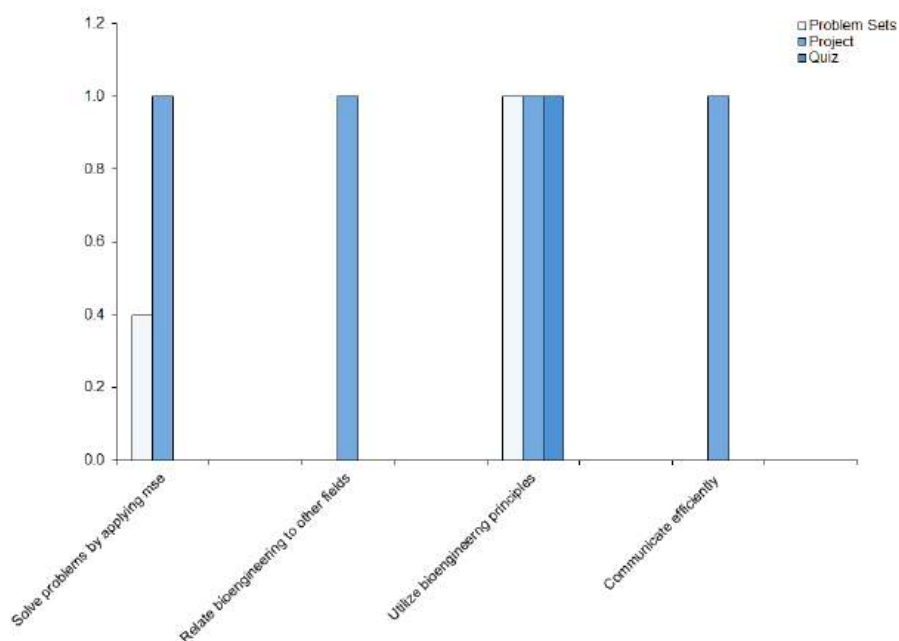


Figure 1. Course Objectives Coverage by Graded Event

3.b. Performance.

Performance on course objectives was assessed using grades taken from those graded events that included significant content in that objective area. The dotted red line shown below indicates the program standard for failure (67%). Some of the cadets still seemed to lack an understanding of how to set up and solve a series of differential or algebraic equations, which hampered their ability to get a numerical solution (i.e. when analyzing the kinetic trends in biological environments) but overall, the class performed relatively well with all cadets passing the course.

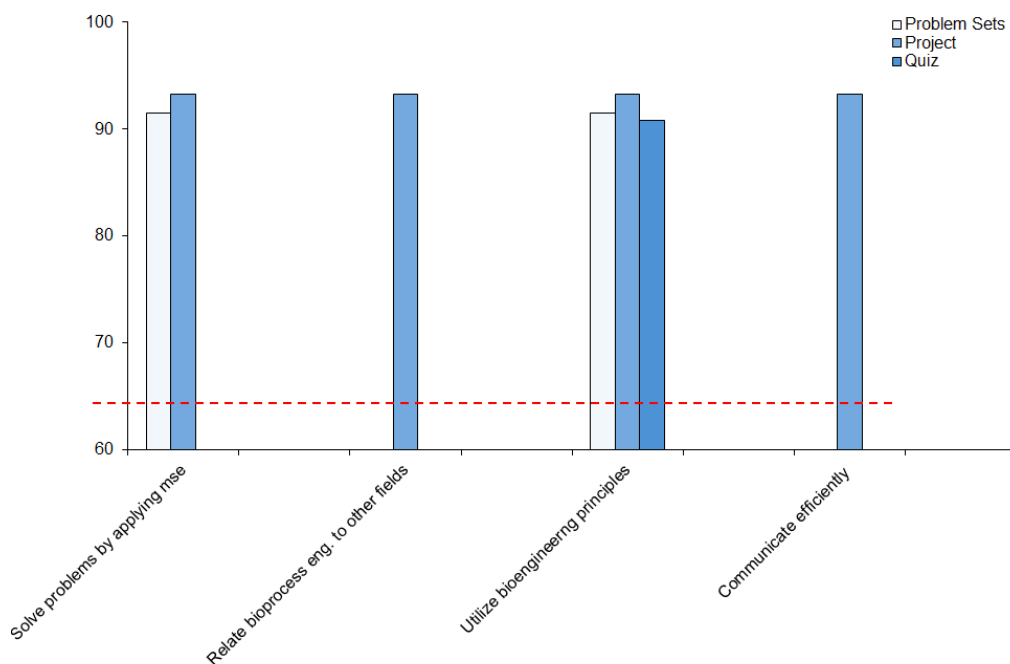


Figure 2. Course Objectives Performance by Graded Event.

4. Survey Questions

4a. Web-based:

The raw data from the web-based survey questions is captured in Section II – Attachment 1. The following graph is a summary of each of the web-based questions. For AY26-2, the number of respondents was 13/17 cadets, or 76.47%. The error bars for the survey appear in the raw data in Attachment 1 and show that the responses for CH450 are within the standard deviations for all departmental, cognitive domain, and academy-wide groups.

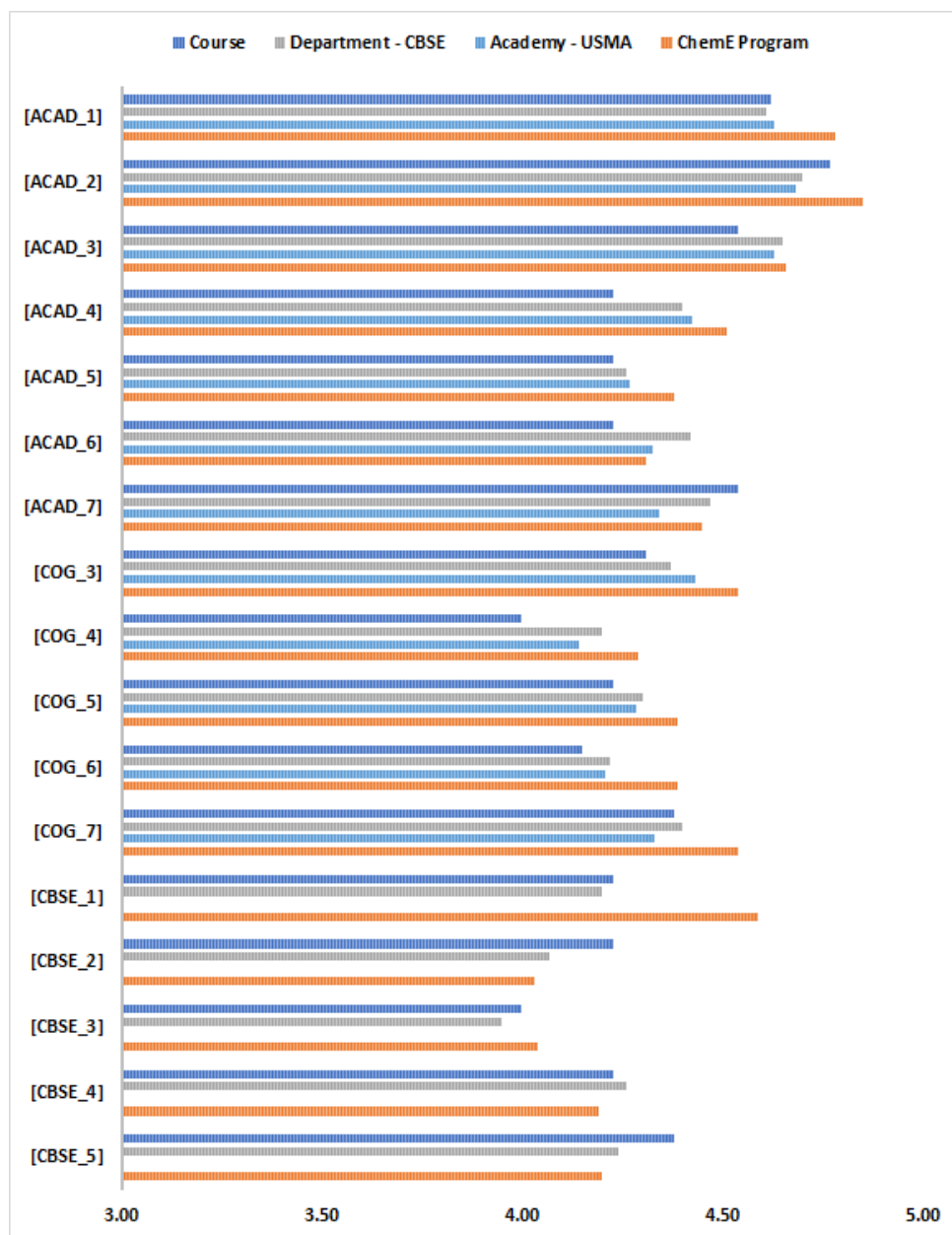


Figure 3. Roll-up of the CoursEval survey assessment results.

4.b. Handwritten Surveys: The student outcomes survey is another method by which we assess student outcomes (SOs). No handwritten survey was conducted on AY26-2 for CH450.

5. Course QPA. The AY26-2 course QPA was 3.55. It is important to note that 8 of 18 cadets majored either in environmental engineering or life science, while the rest of cadet population is from chemical engineering programs. The non-chemical engineering cadet received an average percentage of 90%, showing that he/she possesses enough skills to navigate through the course materials without majoring in chemical engineering program.

6. TEE Grade. No TEE was given for CH450 in AY26-2.

7. Course Processes.

7.a. Textbook. The Enderle Biomedical engineering textbook is relatively suitable for this course. Enderle's book is up to date in terms of math and bioengineering contents. My assessment is that most of the material is presented in a method that is not too difficult for the cadets to follow or understand. Enderle's text suffers from a noticeable drawback wherein core concepts are introduced without sufficient background information, making it challenging for cadets to understand them by purely following the textbook.

The textbook offers some example questions but based heavily on Matlab. Thus, the instructor needs to convert the syntax of coding if he/she wants to use the specific software (i.e. Mathematica) for solving the in-textbook problems. Additional recourses may be needed if the instructors want to show how the core design equations learned in CH450 can be efficiently solved using the computer programs. Again, Mathematica and ChemCad will be good problem candidates for demonstrating coding examples in the class.

7.b. Course Content. The course content and lesson blocks were appropriate for the course. Background reading and homework problems were assigned prior to each class. After each discussion, the instructor would plan to guide the students through an in-class problem or have the students work on an in-class problem.

One of the primary requests from cadets is to demonstrate how to solve complex problems in class. Presenting step-by-step solutions has proven effective in helping cadets understand efficient problem-solving strategies. Sharing partial solutions further enriches their problem-solving skills. Many in-class problems are derived from examples found in various textbooks.

Often, cadets complete their assigned readings before class, but lesson assignments and problem sets were designed to force them into the text. Cadets generally worked problems few days before they were due or, at worst, on the day they were due (by observing the time stamp on the Canvas). Some of them often relied heavily on AI, their smarter colleagues, or any available external sources to solve the problem sets, which potentially degraded the learning of some cadets.

More emphasis can be placed on the usage of computer programs in the class. Since the textbook only offers some coding problems based on Matlab, additional sources might be required to design the coding demo problems. Again, Mathematica and ChemCad can be used to construct coding problems as part of problem set or major graded events.

7.c. Lessons and Labs. The lessons are appropriate for this course. The sequence of lessons is arranged in a way so that cadets can see the course contents moving from molecular scale to reactor scale. No labs were conducted in AY26-2 for CH450.

7.d. Summary of Graded Requirements. The number and type of graded requirements were appropriate for the course. A large bank of problems should be developed so instructors can rotate problems from year to year. Also, I can see that the number of problem sets assigned could be increased to further reinforce the core concepts learned in the class.

7.e. Areas of Special Emphasis. The areas of special emphasis were appropriate for the course.

8. Contribution to Student Outcomes

8.a. Coverage. Contribution to student objectives was assessed using the same method as described in Section II.3. Figure 4 shows coverage of student curriculum outcomes. The spreadsheet containing the data for this analysis is attached to the end of this report, in the section labeled "Section II - Attachment 2 – Assessment Worksheet." The items marked with **red dashed boxes** in Figures 3 designate student directives from Section I.8.

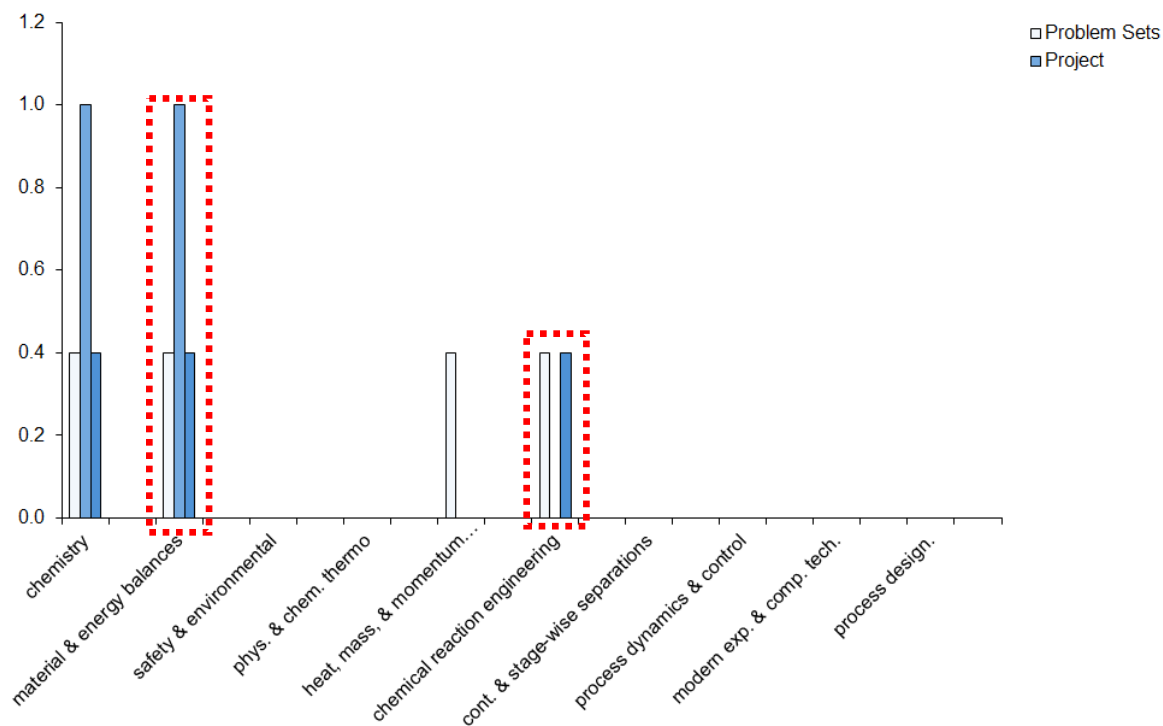


Figure 4. Student Curriculum Outcomes Coverage by Graded Event.

8.b. Performance

Performance on the student outcomes is assessed using the cadet scores from the indicators from the graded events that are outlined above. The results are shown in Figures 5 below. The average for all graded events met or exceeded the 67% standard set forth by USMA for a passing grade. Although none of cadets risked achieving less than 67% in any of area, some of them did struggle with the general chemistry-based problems (i.e., chemical equilibrium and/or kinetics).

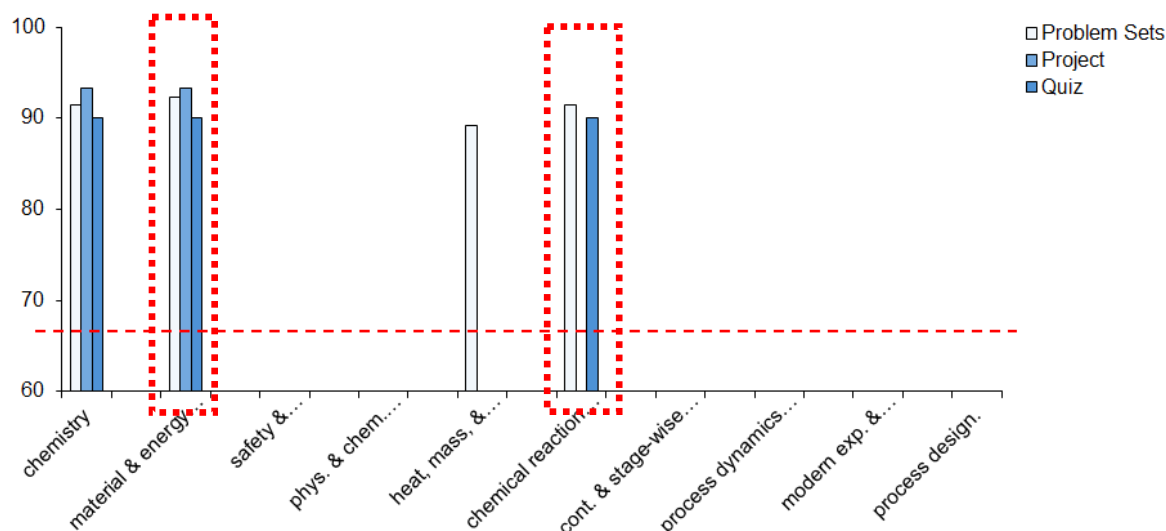


Figure 5. Student Curriculum Outcomes Performance by Graded Event.

9. Resources and Laboratories.

9a. Laboratories. There were no wet labs conducted in this course.

9b. Computer Labs. All class lectures meet in BH630. No computer labs were executed in CH450.

9c. Physical Models & Demos. There are no physical models or demos used in the course.

9d. Technician Support. IT support from CBSE for maintenance of computers in BH630. The IT support from CBSE was adequate.

9e. Supplies. The supplies for the computer lab were adequate (no wet labs were conducted in CH450).

9f. Additional Facilities. None were used - adequate.

9g. Unfunded Requests. None - adequate.

SECTION III: RECOMMENDED CHANGES

1. Red Book Description. No change

2. Enrollment Numbers. Based on the projected enrollment of 18, the course should maintain 1 section and use both associated lab periods physically in the classroom.

3. Course Objectives. No change.

3.a. Coverage. To assess and reinforce the cadets' understanding on bioengineering modeling, multiple in-class and after-class problems can be further incorporated to show how some critical mathematical skills (i.e., the ordinary differential equations (ODEs) can be coupled with the biological phenomena. One lesson needs to be invested to reemphasize the knowledge associated with kinetics and chemical equilibria introduced in general chemistry since some of the problems that cadets struggled the most is closely related to chemical equilibrium and kinetics. Instructor points were adapted to reflect the cadets' contribution towards in-class worksheets during the class.

3.b. Performance. No change.

4. Survey Questions. No change.

4.a. Survey Results. No change.

4.b. Survey Freeform Comments. Continue to include these as options.

5. Course QPA. No change.

6. TEE Grade. No TEE was given in CH450 for AY25-2.

7. Course Process

7.a. Current Textbook. Use of *Enderle, John, and Joseph Bronzino. Introduction to biomedical engineering. Academic press, 2011* is a sustain. The proposed textbooks are comprehensive texts for biomedical engineering and are widely used in the engineering community.

7.b. Course Content. To assess and reinforce the cadets' understanding on bioengineering, multiple in-class and after-class problems were updated and incorporated to show how the biological concepts can be translated into mathematical expressions to get the quantitative results for further analysis. Also, in accordance with the developing the cadet's ability for technical writing, it will be beneficial to include the assignment for writing a technical abstract and developing figures/tables since these are the essential parts of constructing a scientific paper. Continued emphasis on IPRs to provide feedback for cadets on their assignments for development is always important for any type of project-based assignments. The capstone experiences will eventually need to be fully converted into project-based scenarios, so the cadets can spend additional time/efforts working on the engineering processes.

Also, given the usage of generative AI, incorporating an oral examination into either the end-of-semester Capstone Project or the TEE would provide a robust mechanism for assessing cadets' fundamental understanding of bioengineering principles.

7.c. Lessons and Labs. Continue to have cadets work more in-class problems. Working a problem in every lesson is ideal and I recommend continuing to work through the full solution to the instructor problem to demonstrate to the cadets both setting the problem as well as the analytical solution with Mathematica. More time needs to be invested on fundamental reactor mass balances and data analysis. At the same time, cadets' heavy dependency on coding without attempting to access the problems contents need to be alleviated by introducing non-computer based graded events. As mentioned in Section 3.a., revisiting some of major contents from previous course (i.e., general chemistry) may be needed to reinforce the cadets' understanding of kinetic and chemical equilibrium in biological systems.

7.d. Summary of Graded Requirements. Problem sets should be rewritten with new questions each year, as cadets may have access to solutions from previous years or online platforms and the generative AI that provide direct answers to textbook problems. Less-motivated cadets may rely heavily on these resources, often resorting to simply copying existing solutions. Therefore, it is strongly recommended to regularly revise and update major components of in-class quizzes and problem sets to maintain academic integrity and ensure meaningful learning.

7.e. Areas of Special Emphasis. No change.

8. Contribution to Student Outcomes

8.a. Coverage. No change.

8.b. Performance. No change.

9. Resources and Laboratories.

9a. Laboratories. No change.

9b. Computer Labs. No change.

9c. Physical Models & Demos. No change.

9d. Technician Support. No change.

9e. Supplies. No change.

9f. Additional Facilities. No change.

9g. Unfunded Requests. No change.

SECTION IV: GRADED EVENTS AND SUPPLEMENTAL MATERIAL

- 1. Approved Solutions to all WPRS, Quizzes, and Labs.** Digital records are in the CH450 SharePoint graded events directory.
- 2. All supplemental material provided to cadets not included in their book issue purchase.** Digital records are in the CH450 directory.
- 3. A listing of any other resource that provided significant instructional material for cadet use. (Web sites, shared programs, etc.)**

None for CH450.

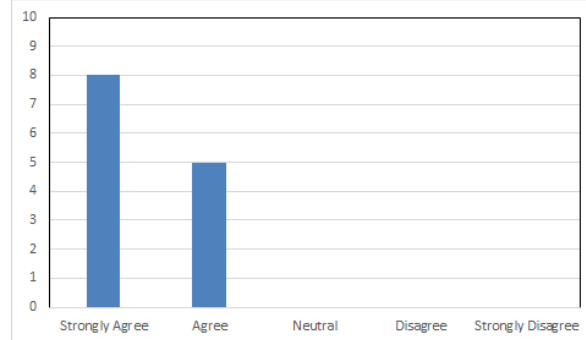
4. Other historical information

4.a. Course Director's Notes

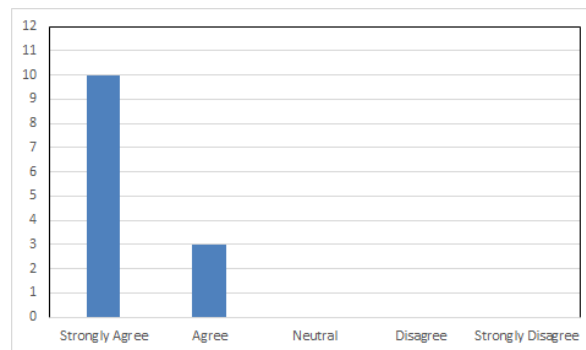
See course director's notebook.

SECTION II: ATTACHMENT 1 – WEB-BASED SURVEY RAW DATA

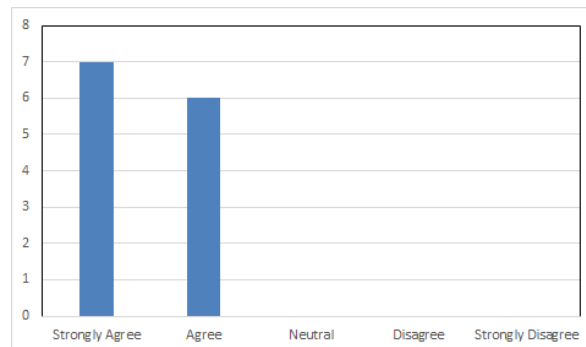
[ACAD_01]: My instructor cared about my learning in this course:



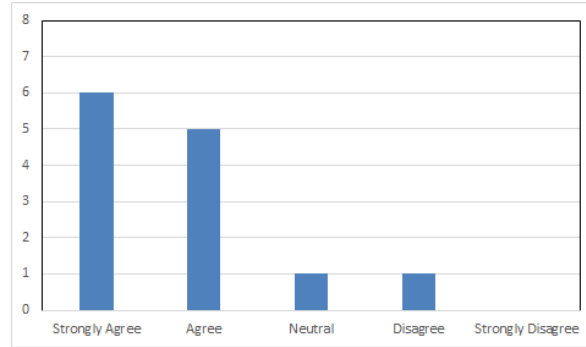
[ACAD_02]: My instructor demonstrated respect for cadets as individuals:



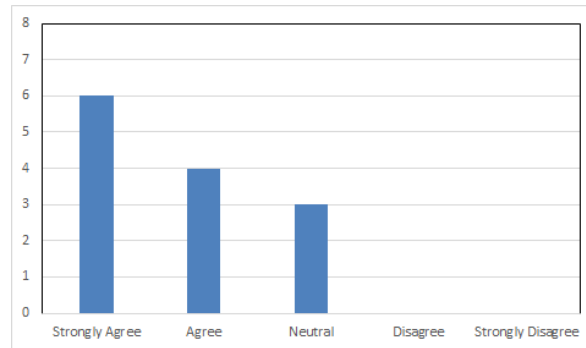
[ACAD_03]: This instructor encouraged students to be responsible for their own learning:



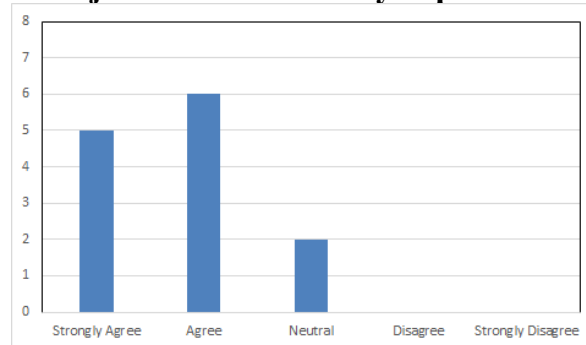
[ACAD_04]: The instructor provided effective feedback:



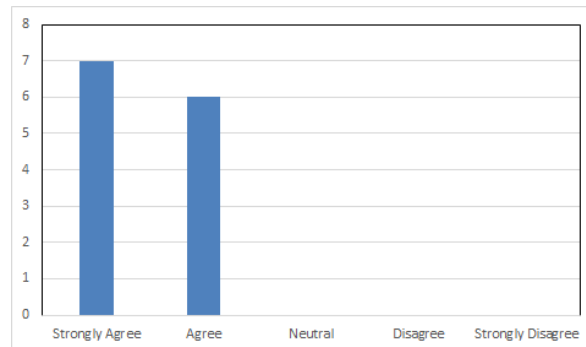
[ACAD_05]: My fellow students contributed to my learning in this course:



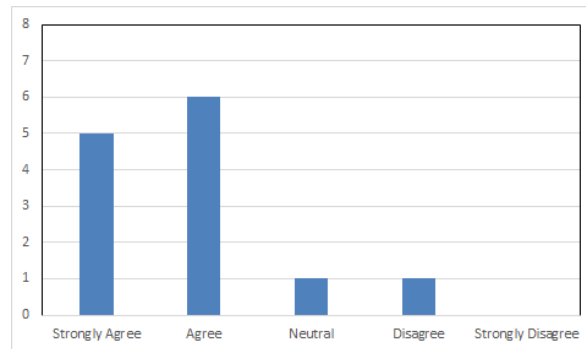
[ACAD_06]: The course objectives were clearly explained:



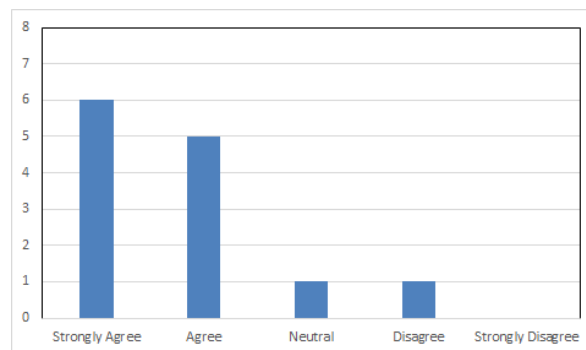
[ACAD_07]: I believe the grading was fair:



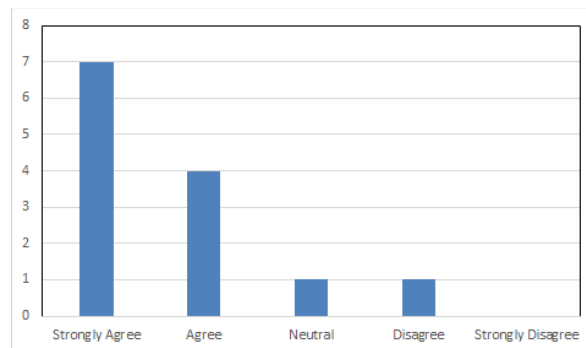
[COG_01]: This instructor used effective techniques for learning, both in class and for out of class assignments:



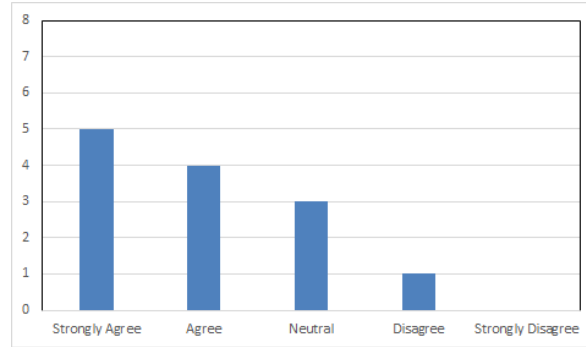
[COG_02]: This instructor stimulated my thinking:



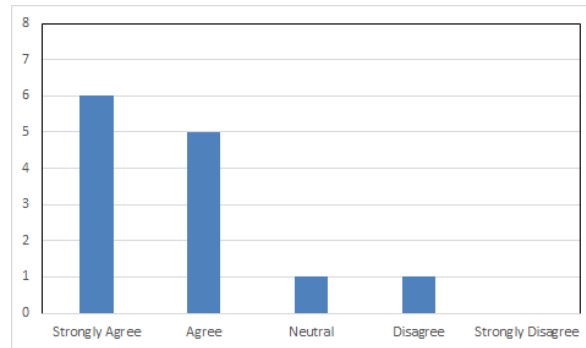
[COG_03]: This instructor encouraged students to think creatively and take intellectual risks



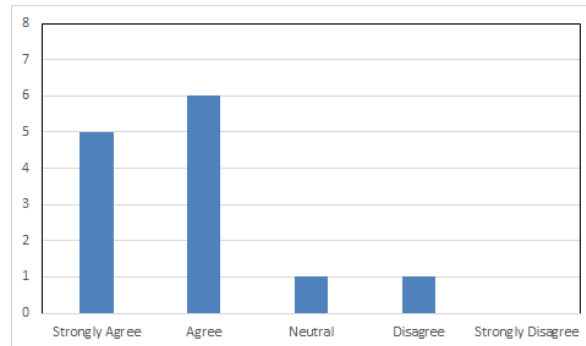
[COG_04]: My motivation to learn and to continue learning has increased because of this course:



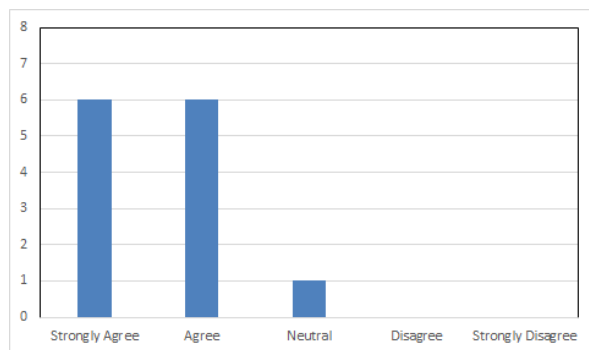
[COG_05]: In this course, my critical thinking ability increased:



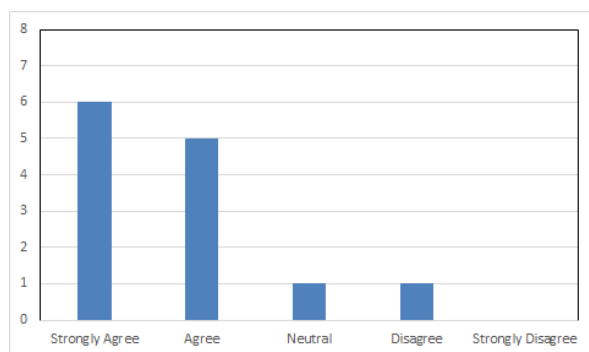
[COG_06]: In this course, my ability to think creatively and take intellectual risks increased:



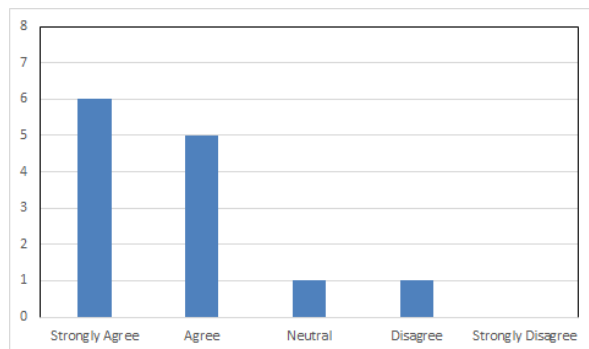
[COG_07]: The course materials were relevant and had a positive contribution to my learning:



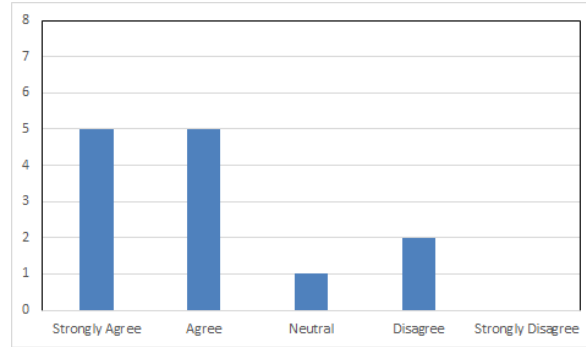
[CBSE_1]: Consider the departments responsible for the courses which you have taken this semester. Relative to other departments, my experience with the Department of Chemical and Biological Science and Engineering was very satisfying:



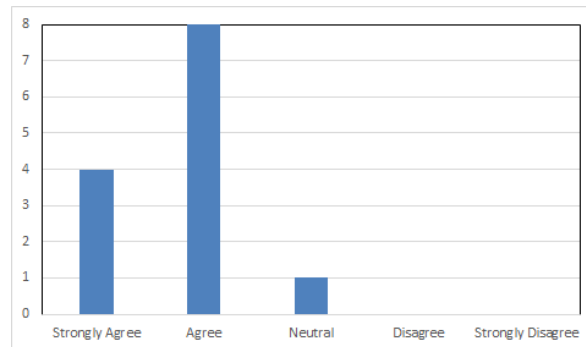
[CBSE_2]: The laboratory program was useful in supporting, reinforcing, or augmenting my understanding of course content:



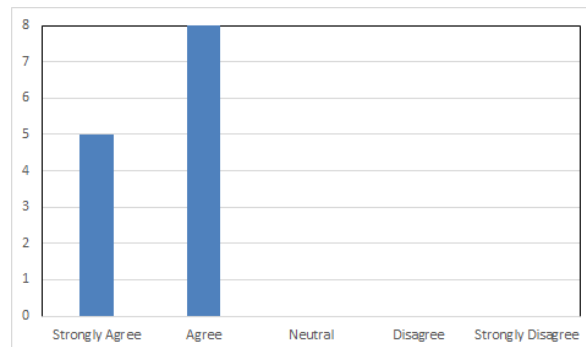
[CBSE_3]: The course text and references were very helpful in my gaining understanding of the course material:



[CBSE_4]: The Course Goals and Lesson Objectives for this course gave me a clear understanding of what was expected of me in order to do well in this course:



[CBSE_5]: The WPRs, block exams, and quizzes in this course were a reasonable evaluation of my understanding of the course material:



SECTION II – ATTACHMENT 2

HANDWRITTEN SURVEY YEARLY COMPARISON

SECTION II – ATTACHMENT 3
STUDENT OUTCOMES WORKSHEETS

Coverage	PS1	PS2	PS3	PS4	PS5	PS	Capstone Present	Capstone Paper	Project	Quiz1	Quiz2	Quiz3	Quiz4	Quiz5	Quiz
Student Outcomes (Outcomes are abbreviated)	← Categories Carried Forward from Grades Spreadsheet														
On completion of the program, our students have demonstrated that they can:															
id, form & solve by appl mse	1	1	1	1	1	1.0	0	0	0.0	1	1	1	1	1	1.0
design to meet needs	0	0	0	0	0	0.0	0	0	0.0	0	0	0	0	0	0.0
communicate	0	0	0	0	0	0.0	1	1	1.0	0	0	0	0	0	0.0
professional ethical impact	0	0	0	0	0	0.0	0	0	0.0	0	0	0	0	0	0.0
function on a team	0	0	0	0	0	0.0	0	0	0.0	0	0	0	0	0	0.0
experiments	0	0	0	0	0	0.0	0	0	0.0	0	0	0	0	0	0.0
new knowledge	0	0	0	0	0	0.0	0	0	0.0	0	0	0	0	0	0.0
The program provides a thorough grounding and working knowledge of:															
chemistry	0	0	1	1	0	0.4	1	1	1.0	0	0	1	1	0	0.4
material & energy balances	0	0	0	1	1	0.4	1	1	1.0	0	0	1	1	0	0.4
safety & environmental	0	0	0	0	0	0.0	0	0	0.0	0	0	0	0	0	0.0
phys. & chem. thermo	0	0	0	0	0	0.0	0	0	0.0	0	0	0	0	0	0.0
heat, mass, & momentum transfer.	0	0	1	0	1	0.4	0	0	0.0	0	0	0	0	0	0.0

chemical reaction engineering	0	0	1	1	0	0.4	0	0	0.0	0	0	1	1	0	0.4
cont. & stage-wise separations	0	0	0	0	0	0.0	0	0	0.0	0	0	0	0	0	0.0
process dynamics & control	0	0	0	0	0	0.0	0	0	0.0	0	0	0	0	0	0.0
modern exp. & comp. tech.	0	0	0	0	0	0.0	0	0	0.0	0	0	0	0	0	0.0
process design.	0	0	0	0	0	0.0	0	0	0.0	0	0	0	0	0	0.0
Course Objectives															
Solve problems by applying mse	0	0	1	1	0	0.4	1	1	1.0	0	0	0	0	0	0.0
Relate bioengineering to other fields	0	0	0	0	0	0.0	1	1	1.0	0	0	0	0	0	0.0
Utilize bioengineering principles	1	1	1	1	1	1.0	1	1	1.0	1	1	1	1	1	1.0
Communicate efficiently	0	0	0	0	0	0.0	1	1	1.0	0	0	0	0	0	0.0
Raw Grade Data															
Average out of 100	← Data Carried Forward from Grades Spreadsheet →														
	95	89	88	95	90	92	93	93	93	93	89	89	91	92	91

Performance	PS1	PS2	PS3	PS4	PS5	PS	Capstone Presenta	Capstone Paper	Project	Quiz1	Quiz2	Quiz3	Quiz4	Quiz5	Quiz
Student Outcomes															
(Outcomes are abbreviated)															
On completion of the program, our students have demonstrated that they can:															
id, form & solve by appl mse	95	89	88	95	90	92	0	0	0	93	89	89	91	92	91
design to meet needs	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
communicate	0	0	0	0	0	0	93	93	93	0	0	0	0	0	0
professional ethical impact	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
function on a team	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
experiments	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
new knowledge	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
The program provides a thorough grounding and working knowledge of:															

chemistry	0	0	88	95	0	91			93	93	93		0	0	89	91	0	90
material & energy balances	0	0	0	95	90	92			93	93	93		0	0	89	91	0	90
safety & environmental	0	0	0	0	0	0			0	0	0		0	0	0	0	0	0
phys. & chem. thermo	0	0	0	0	0	0			0	0	0		0	0	0	0	0	0
heat, mass, & momentum transfer.	0	0	88	0	90	89			0	0	0		0	0	0	0	0	0
chemical reaction engineering	0	0	88	95	0	91			0	0	0		0	0	89	91	0	90
cont. & stage-wise separations	0	0	0	0	0	0			0	0	0		0	0	0	0	0	0
process dynamics & control.	0	0	0	0	0	0			0	0	0		0	0	0	0	0	0
modern exp. & comp. tech.	0	0	0	0	0	0			0	0	0		0	0	0	0	0	0
process design.	0	0	0	0	0	0			0	0	0		0	0	0	0	0	0
Course Objectives																		
Solve problems by applying mse	0	0	88	95	0	91			93	93	93		0	0	0	0	0	0
Relate bioprocess eng. to other fields	0	0	0	0	0	0			93	93	93		0	0	0	0	0	0
Utilize bioengineerng principles	95	89	88	95	90	92			93	93	93		93	89	89	91	92	91
Communicate efficiently	0	0	0	0	0	0			93	93	93		0	0	0	0	0	0



TAB B AY2024 COURSE ASSESSMENT

Department of Chemistry and Life Science
United States Military Academy
West Point, New York 10996

MADN-CHM-LS

3 June 2024

MEMORANDUM FOR Chemical Engineering Program Director, Department of Chemistry and Life Science, United States Military Academy, West Point, New York 10996

SUBJECT: CH450 Chemical Reaction Engineering Course Assessment (AY24-2)

1. The course assessment meeting for CH450 was conducted on 3 June 2024. Attendees: Dr. Simuck F. Yuk, Dr. Enoch A. Nagelli, Dr. Andrew Biaglow, COL Corey James, and LTC Samuel V. Cowart. The topics covered included: 1) revision of graded events; 2) synchronization of CH300 and CH350 concepts with the model-based analysis in CH450. Summary of recommended changes for AY25 are included in Section III.
2. The CH450 course assessment in accordance with the course director handbook from AY24 is attached.
3. POC is the undersigned.

CF: Dr. Biaglow, Dr. Nagelli, LTC James,
LTC Cowart

Dr. Simuck F. Yuk
Assistant Professor, Chemical Engineering
CH450 Course Director

SECTION I: COURSE DESCRIPTION

1. Current Red Book Description.

CH450 (Version: 2019 2) COURSE DETAILS

COURSE	TITLE	EFF YEAR	EFF TERM	CREDIT HOURS
CH450	BIOENGINEERING MDLING & ANLSIS	2019	2	3.0 (BS=0.0, ET=3.0, MA=0.0)

SCOPE

This course provides a broad understanding of bioengineering disciplines to include biomechanics, biomaterials, tissue engineering, biocatalysis, biochemical engineering, and biosensors. Fundamental concepts of molecular kinetics, thermodynamics, and mass transport are applied in problem sets in each bioengineering sub-discipline and capstone design project providing students the opportunity for modeling, analysis, and design from the biomolecular to physiological length scale and across multiple time scales. Modeling software such as MATLAB and Mathematica are extensively used.

LESSONS: 30 @ 75 min (2.000 Att/wk) **LABS:** 0 @ 0 min

SPECIAL REQUIREMENTS:

None

CH350 COURSE REQUISITES

TYPE	COURSE	EFF YEAR	EFF TERM	TRACK	RED BOOK FLG
PRE-REQUISITE	CH102	2019	2	10	Y
	MA205	2017	1	9	Y
	MA255	2016	2	10	Y
	PH202	2023	1	8	Y
	PH252	2023	1	1	Y

2. Enrollment Numbers.

- AY 2024-2: 18
- AY 2025-2: 15 (projected)
- AY 2026-2: 15 (projected)
- AY 2027-2: 18 (projected)

3. Current Course Objectives.

This course provides a broad understanding of bioengineering disciplines to include biomechanics, biomaterials, tissue engineering, biocatalysis, biochemical engineering, and biosensors. Specifically, the objectives of the course are:

- 1) Understand the broad meaning of the term "bioengineering" and the interface between research, engineering, and clinical fields.
- 2) Synthesize math, science, and engineering concepts from the Core Sequence and major courses in a biological/bioengineering context.
- 3) Understand, apply, and manipulate models for bioengineering design.
- 4) Reinforce and strengthen conceptual and practical understanding of fundamental thermodynamics, kinetics, and mass transport.
- 5) Further develop critical thinking skills using journal articles.
- 6) Develop oral and written communication skills and continue to develop cadets' ability to prepare technical reports.

4. Survey Questions. Below are questions to the web-based survey of questions asked to CH450 cadets at the end of the AY24-2 semester:

Academy

Q1. My fellow students contributed to my learning in this course.

Q2. My motivation to learn and to continue learning has increased because of this course.

Q3. My personal schedule allows me enough time to reflect on the material I have learned in class.

Q4. My personal schedule allows me enough time to adequately prepare for my optimum academic performance.

Academy - I

Q5. This instructor encouraged students to be responsible for their own learning.

Q6. This instructor used effective techniques for learning, both in class and for out-of-class assignments.

Q7. My instructor cared about my learning in this course.

Q8. My instructor demonstrated respect for cadets as individuals.

Cognitive Domain

Q9. In this course, my critical thinking ability increased.

Q10. The homework assignments, papers, and projects in this course could be completed within the USMA time guideline of a 2:1 ratio of out-of-class time versus in-class time.

Q11. The classroom environment (e.g., desk setup, boards, technology, lights, etc.) had a positive impact on my ability to learn.

Q12. In this course, my ability to think creatively and take intellectual risks increased.

Q13. This instructor stimulated my thinking.

Q14. This instructor encouraged students to think creatively and take intellectual risks.

Department

Q15. Consider the departments responsible for the courses which you have taken this semester. Relative to other departments, my experience with the Department of Chemistry and Life Sciences was very satisfying.

Q16. The laboratory program was useful in supporting, reinforcing, or augmenting my understanding of course content.

Q17. The course text and references were very helpful in my gaining understanding of the course material.

Q18. The Course Goals and Lesson Objectives for this course gave me a clear understanding of what was expected of me to do well in this course.

Q19. The WPRs, block exams, and quizzes in this course were a reasonable evaluation of my understanding of the course material.

5. Course GPA. The GPA for the course from the last 7 years follows below.

AY24-2 (Spring 2024) 3.61

6. TEE Grade.

No TEE implemented in CH450 for AY24-2.

7.a. Current Textbook.

Enderle, John, and Joseph Bronzino. Introduction to biomedical engineering. Academic press, 2011, ISBN: 978-0-12-374979-6.

7.b. Course Content.

Part I: Biomechanics

Part II: Biomaterials

Part III: Tissue Engineering

Part IV: Biochemical Reactions and Enzyme Kinetics

Part V: Biosensors and Bioimaging

7.c. Lesson Sequence

LESSON	TOPICS
Part I Biomechanics	
1	Introduction & Modeling
2	Biomechanics I
3	Biomechanics II
4	Biomechanics III
5	Modeling: Biomechanics
Part II Biomaterials	
6	Biomaterials I
7	Biomaterials II
8	Biomaterials III
9	Biomaterials IV
10	Modeling: Biomaterials
Part III Tissue Engineering	
11	Tissue Engineering I
12	Tissue Engineering II
13	Tissue Engineering III
14	Tissue Engineering IV
15	Modeling: Tissue Engineering
Part IV Biochemical Reactions and Enzyme Kinetics	
16	Compartment Modeling
17	Biochemical Enzyme Kinetics I
18	Biochemical Enzyme Kinetics II
19	Biochemical Enzyme Kinetics III
20	Biochemical Enzyme Kinetics IV
21	Modeling: Biochemical Enzyme Kinetics
Part V Biosensors and Bioimaging	
22	Biomedical Sensors I
23	Biomedical Sensors II
24	Special Topic: Guest Lecture
25	Bio-Imaging I
26	Bio-Imaging II
27	Modeling: Biomedical Sensors
28	Course Review
29	Capstone Presentations I
30	Capstone Presentations II

7.d. Summary of Graded Requirements

Requirement	Number	Points	Total	%
Problem Sets	5	100	500	50%
Quizzes	5	30	150	15%
Capstone Presentation/IPR	1	100	100	10%
Capstone Paper	1	150	150	15%
Instructor Points	---	100	100	10%
TOTAL			1000	100%

7.e. Areas of Special Emphasis.

Course Director's Area of Special Emphasis

My primary area of interest was Bioengineering modeling and analysis. This course emphasized the development of mathematical models and how these models can be applied to analyze the quantitative results from various biological systems. A large portion of knowledge was geared toward translating biological concepts into mathematical forms and creating working models. I wanted cadets to understand how to develop reasonable governing equations by assuming the appropriate initial and boundary conditions, so they can obtain and analyze the quantitative results as an engineer.

My second area of special emphasis was on the design of a fermenter utilizing biological components. The problems demonstrated by the instructor in class centered around designing a reactor system that integrates material balance across cell, substrate, and production concentrations. Throughout the course, various bioreactor designs were introduced, highlighting the distinct reactor conditions present in batch and continuous cultivation.

My third area of special emphasis was on the ability to communicate effectively. As part of the capstone project, cadets are required to individually submit a technical abstract and subsequently produce a final research paper. Part of the grade on professionalism of the written report is devoted to their communication skills. Cadets are evaluated on their technical writing and oral presentation skills, along with the feedback on different aspects of capstone projects.

My fourth area of special emphasis was on the use of modern computing techniques. Some of the kinetic problems in this course involved higher level calculus that required the use of an ODE solver, i.e., Mathematica, to project the kinetic trends in efficient manners. Several problems, i.e., solving Michaelis Menten kinetics for different species presented in the chemostat, were introduced with the emphasis of using computing codes to approach the numerical solutions.

8. Contribution to Student Outcomes

The following list contains the Chemical Engineering Student Outcomes. The phrases in **bold-faced red font** are the topics that this course is specifically asked to accomplish by the program. These bullets are also indicated with **red dashed boxes** in Section II.8

The Chemical Engineering Program provides the graduate with a thorough grounding and working knowledge of ...

- General, organic, and physical chemistry.
- **Material and energy balances as well as safety and environmental factors.**

- Thermodynamics of physical and chemical equilibria.
- Heat, mass, and momentum transfer.
- **Chemical reaction engineering.**
- Continuous and staged separation operations.
- Process dynamics and control.
- Modern experimental and computing techniques.
- Process design.

9. Resources and Laboratories.

9a. Laboratories. There were no wet labs conducted in this course.

9b. Computer Labs. All class lectures meet in BH630. No labs were executed in CH450.

9c. Physical Models & Demos. There are no physical models or demos used in the course.

9d. Technician Support. IT support from the CLS IT office for maintenance of computers in BH630 was timely and addressed any issues that arose. In future considerations for distance learning, a scanner, Epson elmo, and a second monitor are important to have at designated location of teaching.

9e. Supplies. Mathematica 11 and Microsoft Office (Word, PowerPoint, Excel, and Explorer).

9f. Additional Facilities. None.

9g. Unfunded Requests. None.

10. Recommendations from AY23-2 and Implementation in AY24-2:

<i>Section I.</i>	<i>Recommendation (AY23-2)</i>	<i>Implementation (AY24-2)</i>
1. Redbook	No change.	Implemented as recommended (IAR)
2. Enrollment	No change.	No change.
3. Course Objectives	No change.	No change.
3.A. Coverage	No change.	The synchronization between CH300, CH350, and CH450 will eventually needed if the program were to create the engineering sequence for these bioengineering electives.
3.B. Performance	No change.	No change.
4. Survey Questions	No change.	The end-of-course survey administered through CoursEval had a response rate of 40%. The extra instructor points can

		be offered to encourage cadets' participating in survey.
5. Course QPA	No change.	No change.
6. TEE Grades	No change.	No change.
7. Course Processes 7.A. Textbook	No change.	No change.
7.B. Course Content	No change.	No change.
7.C. Lessons and Labs	No change.	No change.
7.D. Graded Events	No change.	A new type of problem was introduced as part of in-class activities, so instructors can assess the cadets' understanding of learning objectives. The introduction of in-class activities also facilitates the cadets' coding skills in Mathematica.
7.E. Areas of Special Emphasis	No change.	IAR
8. Contribution to Student Outcomes 8.A. Coverage	No change.	IAR
8.B. Performance	No change.	IAR
9. Resources and Laboratories 9.A. Laboratories	No change.	IAR
9.B. Computer Labs	No change.	IAR
9.C. Physical Models & Demos	No change.	IAR
9.D. Technician Support	No change.	IAR
9.E. Supplies	No change.	IAR
9.F. Additional Facilities	No change.	IAR

9.G. Unfunded Requests	No change.	IAR
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SECTION II - COURSE ASSESSMENT

1. Red Book Description. The Red Book description was followed verbatim.

2. Enrollment. The enrollment in the course was 18 for AY24-2, with the projected enrollment in AY25-2 to be 15. During Term 24-2, no cadets failed the course, and none were at serious risk of failing the course going into the Term End.

3. Course Objectives. The course objectives were covered thoroughly and achieved with proficiency. The course objectives cover the body of knowledge appropriately. The course objectives lend themselves to assessment.

3.b. Coverage

Coverage of course objectives is presented in Figure 1. As in the other courses in this program, coverage is assessed by the course director, who assigns a 1 or a 0 to a graded event or portion of a graded event. A score of 1 on a graded event or a portion of a graded event designates significant coverage, while a 0 designates an insignificant amount or no coverage. Coverage scores are then averaged across all similar graded events in the course (problem sets, exams, etc.). A score close to 1 indicates a relatively large portion of that type of graded event covered that objective. For example, a score of 0.6 on design problems under Objective 1 (mass/energy balances) indicates that 60% or 3 out of 5 of the problem sets contained coverage of that objective. This does not say that the other problem set had zero coverage of that objective, only that it was not counted by the course director for assessment. An objective with multiple bars shows that multiple types of graded events covered that objective. Raw data are included in the spreadsheet attached to the end of this document in Attachment 3. Because of this analysis, we conclude that course objectives were covered.

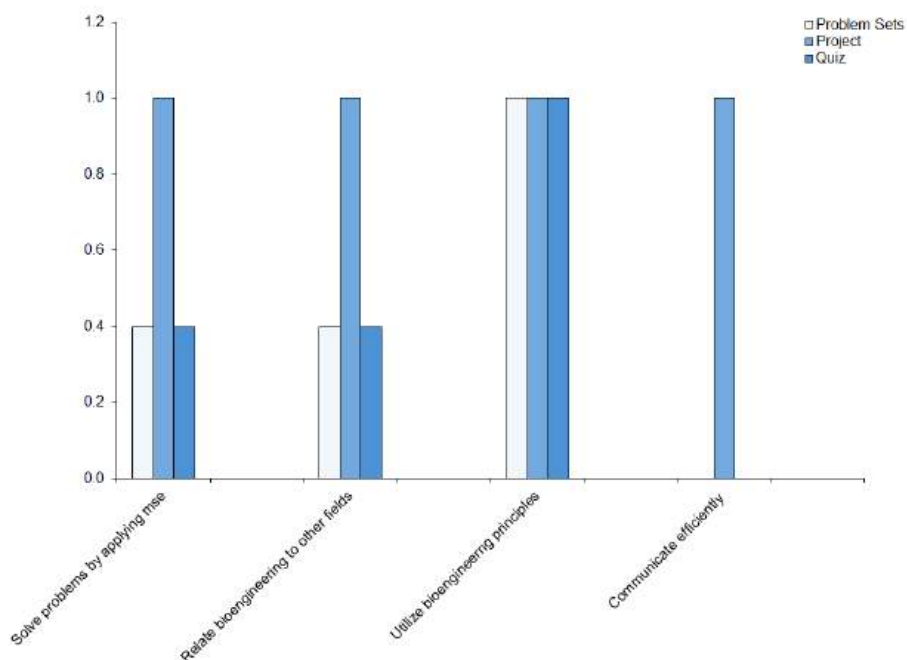


Figure 1. Course Objectives Coverage by Graded Event

3.b. Performance.

Performance on course objectives was assessed using grades taken from those graded events that included significant content in that objective area. The dotted red line shown below indicates the program standard for failure (67%). Some of the cadets still seemed to lack an understanding of how to set up and solve a series of differential or algebraic equations, which hampered their ability to get a numerical solution (i.e. when analyzing the kinetic trends in biological environments) but overall, the class performed relatively well with all cadets passing the course.

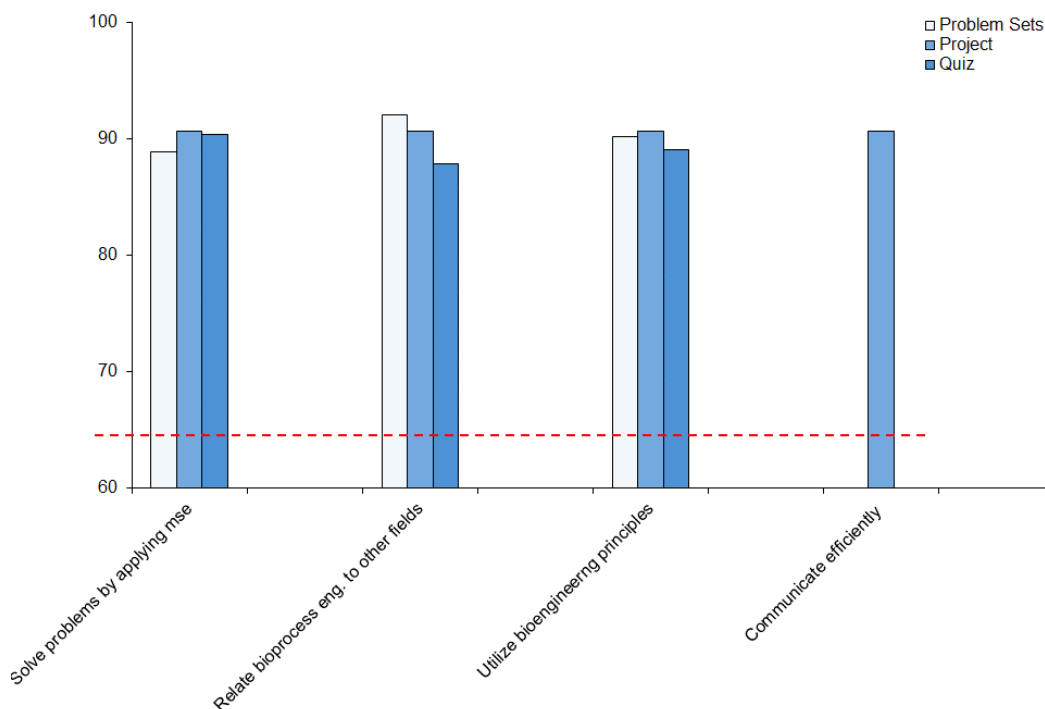


Figure 2. Course Objectives Performance by Graded Event.

4. Survey Questions

4a. Web-based:

The raw data from the web-based survey questions is captured in Section II – Attachment 1. The following graph is a summary of each of the web-based questions. For AY 24-2, the number of respondents was 7/18 cadets, or 40.00%. The error bars for the survey appear in the raw data in Attachment 1 and show that the responses for CH450 are within the standard deviations for all departmental, cognitive domain, and academy-wide groups.

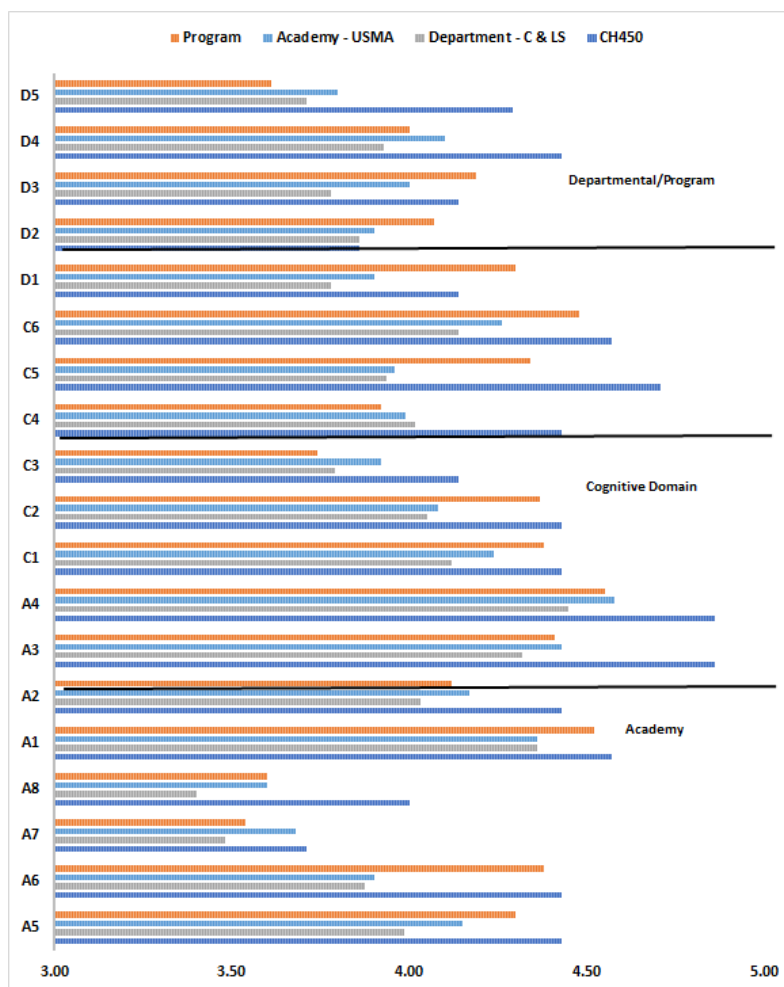


Figure 3. Roll-up of the CourseEval survey assessment results.

4.b. Handwritten Surveys: The student outcomes survey is another method by which we assess student outcomes (SOs). No handwritten survey was conducted in AY24-2 for CH450.

5. Course QPA. The AY24-2 course QPA was 3.61. It is important to note that 4 of 18 cadets majored either in chemistry or life science, while the rest of cadet population is from chemical engineering program. The non-chemical engineering cadet received the average percentage of 91%, showing that he/she possesses enough skills to navigate through the course materials without being majoring in engineering program. Additional data will be needed to make detailed assessment on the impact of non-engineering majors on the cadets' performance in CH450.

6. TEE Grade. No TEE was given for CH450 in AY24-2.

7. Course Processes.

7.a. Textbook. The Enderle Biomedical engineering textbook is relatively suitable for this course. Enderle's book is up to date in terms of math and bioengineering contents. My assessment is that most of the material is presented in a method that is not too difficult for the cadets to follow or understand. Enderle's text suffers from a noticeable drawback wherein core concepts are

introduced without sufficient background information, making it challenging for cadets to understand them by purely following the textbook.

The textbook offers some coding example questions but based heavily on Matlab. Thus, the instructor needs to convert the syntax of coding if he/she wants to use the specific software (i.e. Mathematica) for solving the in-textbook problems. Additional recourses may be needed if the instructors want to show how the core design equations learned in CH450 can be efficiently solved using the computer programs. Again, Mathematica and ChemCad will be good problem candidates for demonstrating coding examples in the class.

7.b. Course Content. The course content and lesson blocks were appropriate for the course. Background reading and homework problems were assigned prior to each class. After each discussion, the instructor would plan to guide the students through an in-class problem or have the students work an in-class problem.

One of the primary requests from cadets is to demonstrate how to solve complex problems in class. Presenting step-by-step solutions has proven effective in helping cadets understand efficient problem-solving strategies. Sharing partial solutions further enriches their problem-solving skills. Many in-class problems are derived from examples found in various textbooks.

Often, cadets completed their assigned readings before class, but lesson assignments and problem sets were designed to force them into the text. Cadets generally worked problem few days before they were due or, at worst, on the day they were due (by observing the time stamp on the Canvas). Some of them often relied heavily on AI, their smarter colleagues, or any available external sources to solve the problem sets, which potentially degraded the learning of some cadets.

More emphasis can be placed on the usage of computer programs in the class. Since the textbook only offers some coding problems based on Matlab, additional sources might be required to design the coding demo problems. Again, Mathematica and ChemCad can be used to construct the coding problems as part of problem set or major graded events.

7.c. Lessons and Labs. The lessons are appropriate for this course. The sequence of lessons is arranged in a way so that cadets can see the course contents are moving from molecular scale to reactor scale. No labs were conducted in AY24-2 for CH450.

7.d. Summary of Graded Requirements. The number and type of graded requirements was appropriate for the course. A large bank of problems should be developed so instructors can rotate problems from year to year. Also, I can see that the number of problem sets assigned could be increased to further reinforce the core concepts learned in the class.

7.e. Areas of Special Emphasis. The areas of special emphasis were appropriate for the course.

8. Contribution to Student Outcomes

8.a. Coverage. Contribution to student objectives was assessed using the same method as described in Section II.3. Figure 4 shows coverage of student curriculum outcomes. The spreadsheet containing the data for this analysis is attached to the end of this report, in the section labeled "Section II - Attachment 2 – Assessment Worksheet." The items marked with **red dashed boxes** in Figures 3 designate student directives from Section I.8.

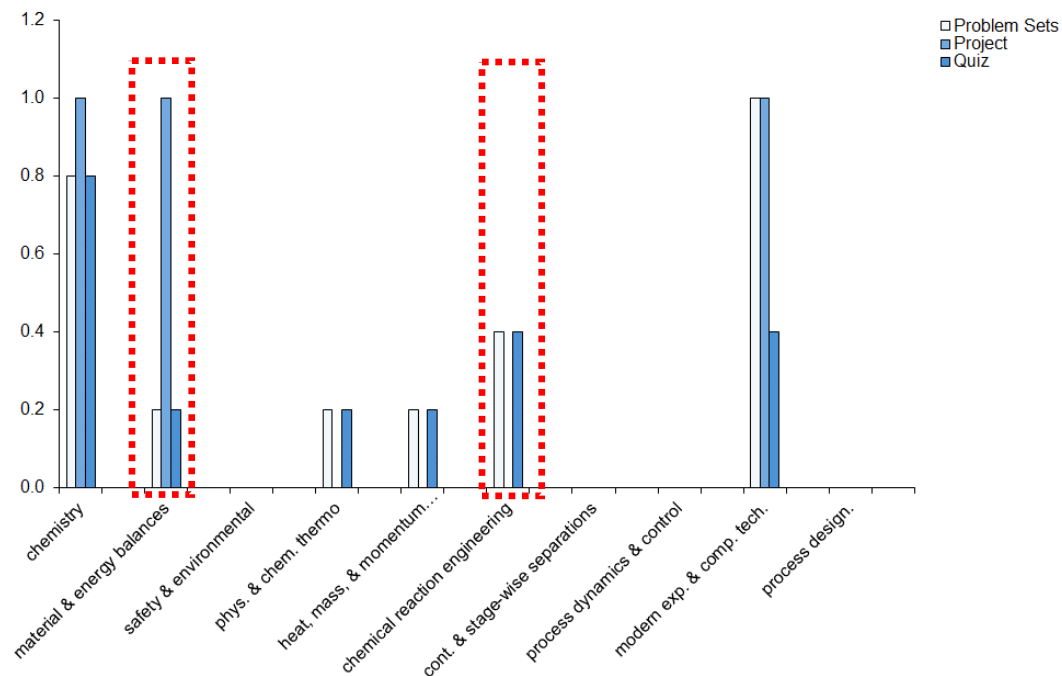


Figure 4. Student Curriculum Outcomes Coverage by Graded Event.

8.b. Performance

Performance on the student outcomes is assessed using the cadet scores from the indicators from the graded events that are outlined above. The results are shown in Figures 5 below. The average for all graded events met or exceeded the 67% standard set forth by USMA for a passing grade. Although none of cadets risked achieving less than 67% in any of area, some of them did struggle with the general chemistry-based problems (i.e., chemical equilibrium and/or kinetics).

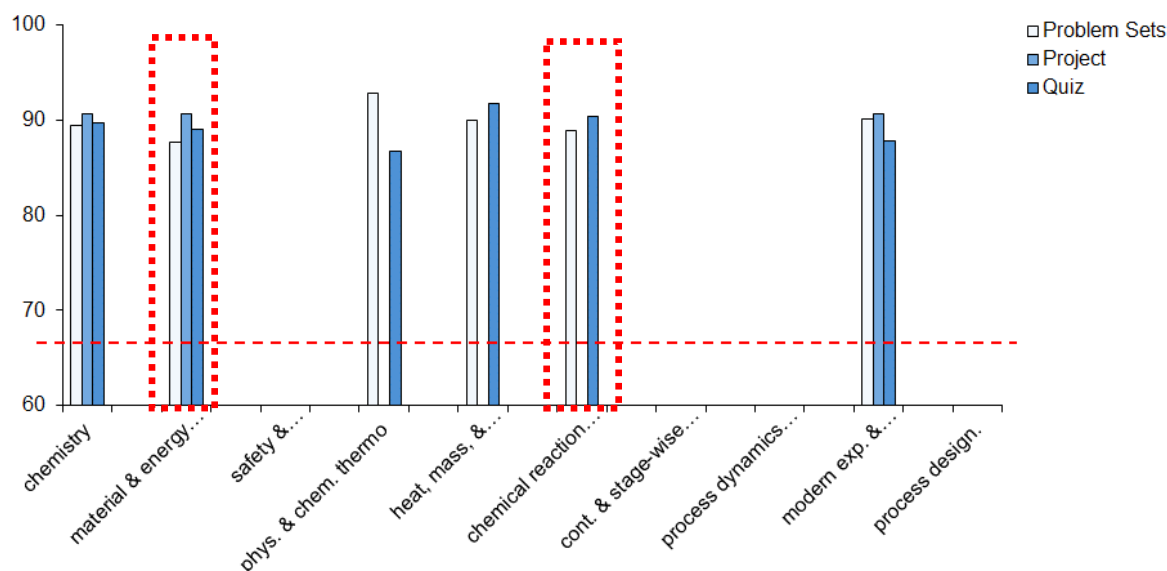


Figure 5. Student Curriculum Outcomes Performance by Graded Event.

9. Resources and Laboratories.

9a. Laboratories. There were no wet labs conducted in this course.

9b. Computer Labs. All class lectures meet in BH630. No computer labs were executed in CH450.

9c. Physical Models & Demos. There are no physical models or demos used in the course.

9d. Technician Support. IT support from CLS for maintenance of computers in BH630. The IT support from CLS was adequate.

9e. Supplies. The supplies for the computer lab were adequate (no wet labs were conducted in CH450).

9f. Additional Facilities. None were used - adequate.

9g. Unfunded Requests. None - adequate.

SECTION III: RECOMMENDED CHANGES

1. Red Book Description. No change

2. Enrollment Numbers. Based on the projected enrollment of 18, the course should maintain 1 section and use both associated lab periods physically in the classroom.

3. Course Objectives. No change.

3.a. Coverage. To assess and reinforce the cadets' understanding on bioengineering modeling, multiple in-class and after-class problems can be further incorporated to show how some of critical mathematical skills (i.e., the ordinary differential equations (ODEs) can be coupled with the biological phenomena. One lesson needs to be invested to reemphasize the knowledges associated with kinetics and chemical equilibria introduced in general chemistry since some of the problems that cadets struggled the most is closely related to chemical equilibrium and kinetics. Instructor points were adapted to reflect the cadets' contribution towards in-class worksheets during the class.

3.b. Performance. No change.

4. Survey Questions. No change.

4.a. Survey Results. No change.

4.b. Survey Freeform Comments. Continue to include these as options.

5. Course QPA. No change.

6. TEE Grade. No TEE was given in CH450 for AY24-2.

7. Course Process

7.a. Current Textbook. Use of *Enderle, John, and Joseph Bronzino. Introduction to biomedical engineering. Academic press, 2011* is a sustain. The proposed textbooks are comprehensive texts for biomedical engineering and are widely used in the engineering community.

7.b. Course Content. To assess and reinforce the cadets' understanding on bioengineering, multiple in-class and after-class problems were incorporated to show how the biological concepts can be translated into mathematical expressions to get the quantitative results for further analysis. Also, in accordance with the developing the cadet's ability for technical writing, it will be beneficial to include the assignment for writing a technical abstract and developing figures/tables since these are the essential parts of constructing a scientific paper. Continued emphasis on IPRs to provide feedback for cadets on their assignments for development is always important for any type of project-based assignments. The capstone experiences will eventually need to be fully converted into the project-based scenarios, so the cadets can spend additional time/efforts working on the engineering processes.

7.c. Lessons and Labs. Continue to have cadets work more in-class problems. Working a problem in every lesson is ideal and I recommend continuing to work through the full solution to the instructor problem to demonstrate to the cadets both setting the problem as well as the analytical solution with Mathematica. More time needs to be invested on fundamental reactor mass balances and data analysis. At the same time, cadets' heavy dependency on coding without attempting to access the problems contents need to be alleviated by introducing non-computer based graded events. As mentioned in Section 3.a., revisiting some of major contents from previous course (i.e., general chemistry) may be needed to reinforce the cadets' understanding of kinetic and chemical equilibrium in biological systems.

7.d. Summary of Graded Requirements. Problem sets need to be rewritten with new problems. Cadets may have access to previous years' solutions and online resources that may have direct answers to textbook questions (e.g., Chegg.com). Some of the less-motivated cadets will do little more than copying the old solutions. As always, rewriting/adjusting the major portions of in-class quizzes and problem sets is strongly recommended for each year.

7.e. Areas of Special Emphasis. No change.

8. Contribution to Student Outcomes

8.a. Coverage. No change.

8.b. Performance. No change.

9. Resources and Laboratories.

9a. Laboratories. No change.

9b. Computer Labs. No change.

9c. Physical Models & Demos. No change.

9d. Technician Support. No change.

9e. Supplies. No change.

9f. Additional Facilities. No change.

9g. Unfunded Requests. No change.

SECTION IV: GRADED EVENTS AND SUPPLEMENTAL MATERIAL

- 1. Approved Solutions to all WPRS, Quizzes, and Labs.** Digital records are in the CH450 SharePoint graded events directory.
- 2. All supplemental material provided to cadets not included in their book issue purchase.** Digital records are in the CH450 directory.
- 3. A listing of any other resource that provided significant instructional material for cadet use. (Web sites, shared programs, etc.)**

None for CH450.

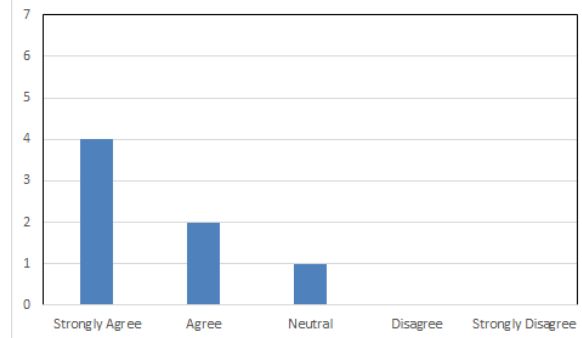
4. Other historical information

4.a. Course Director's Notes

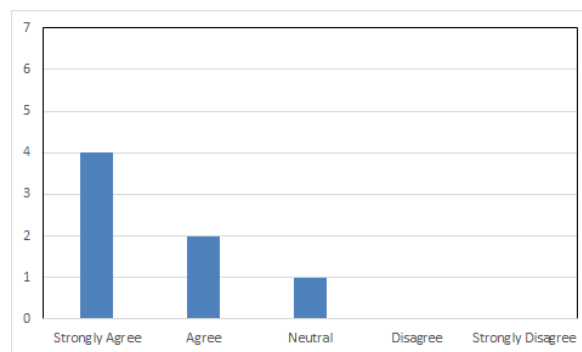
See course director's notebook.

SECTION II: ATTACHMENT 1 – WEB-BASED SURVEY RAW DATA

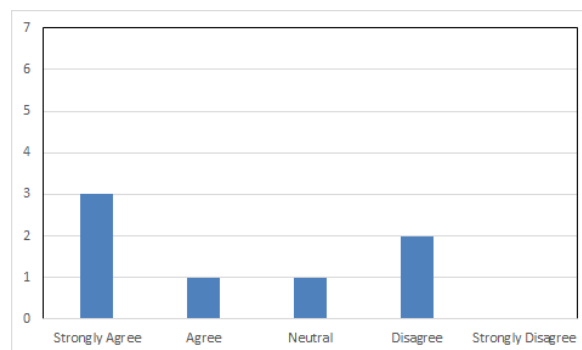
My fellow students contributed to my learning in this course:



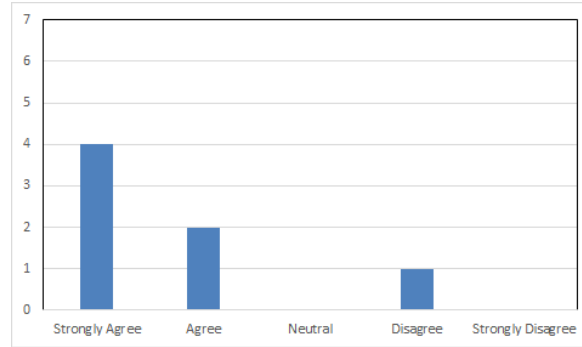
My motivation to learn and to continue learning has increased because of this course:



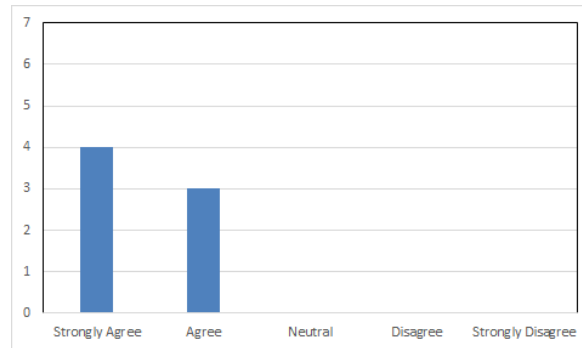
My personal schedule allows me enough time to reflect on the material I have learned in class:



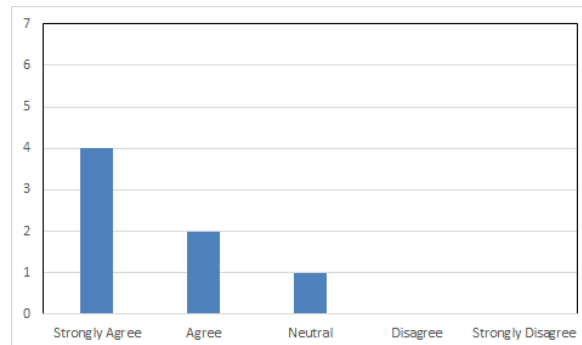
My personal schedule allows me enough time to adequately prepare for my optimum academic performance:



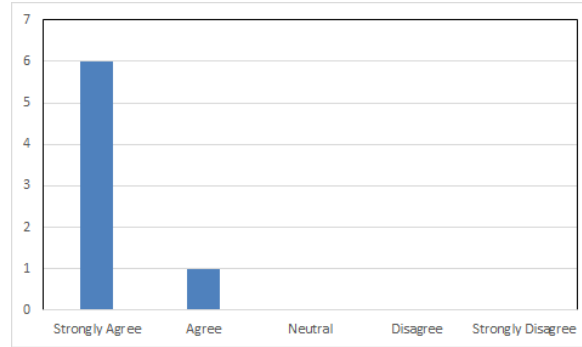
This instructor encouraged students to be responsible for their own learning:



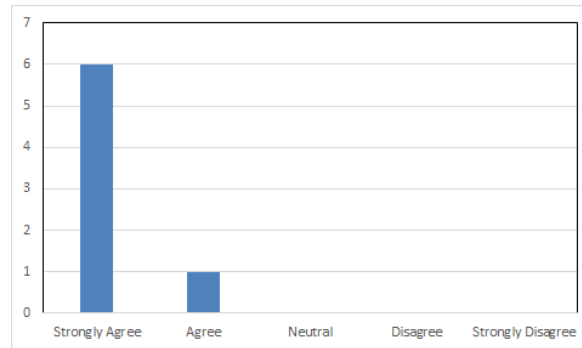
This instructor used effective techniques for learning, both in class and for out of class assignments:



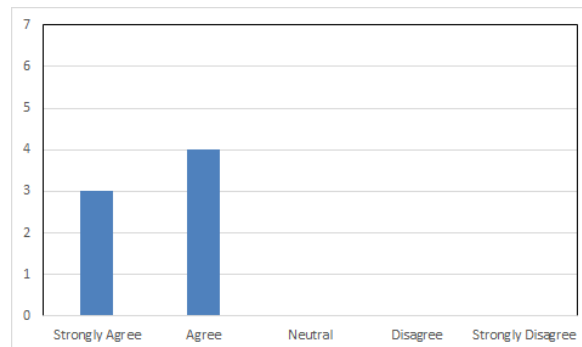
My instructor cared about my learning in this course:



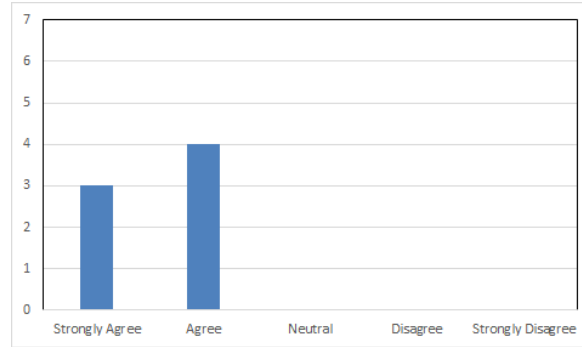
My instructor demonstrated respect for cadets as individuals:



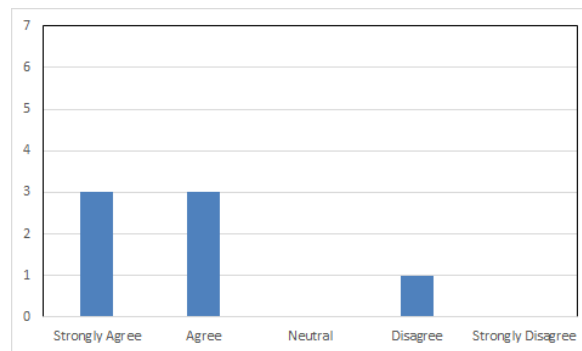
In this course, my critical thinking ability increased:



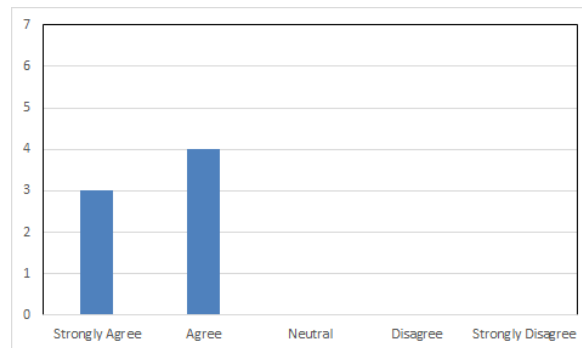
The homework assignments, papers, and projects in this course could be completed within the USMA time guideline of a 2:1 ratio of out-of-class time versus in-class time:



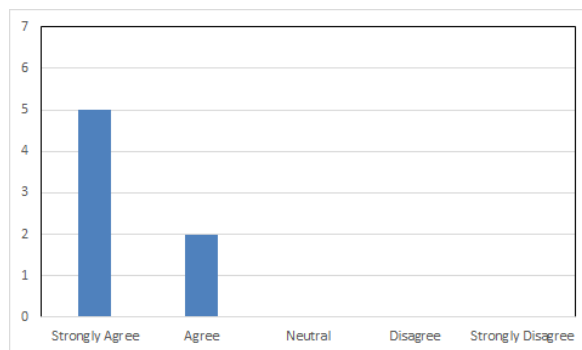
The classroom environment (e.g., desk setup, boards, technology, lights, etc.) had a positive impact on my ability to learn:



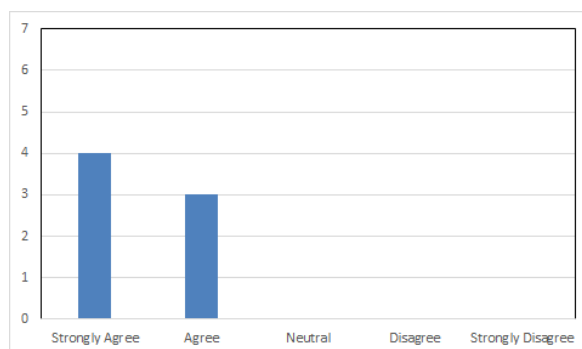
In this course, my ability to think creatively and take intellectual risks increased:



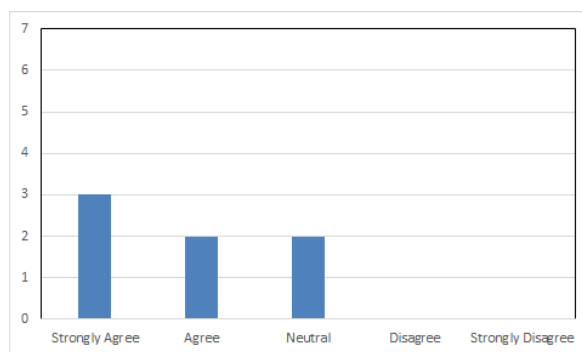
This instructor stimulated my thinking:



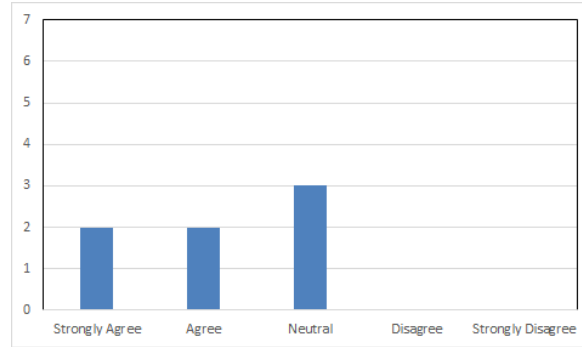
This instructor encouraged students to think creatively and take intellectual risks:



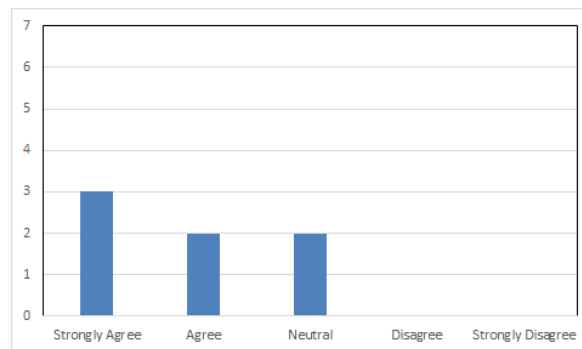
Consider the departments responsible for the courses which you have taken this semester. Relative to other departments, my experience with the Department of Chemistry and Life Sciences was very satisfying:



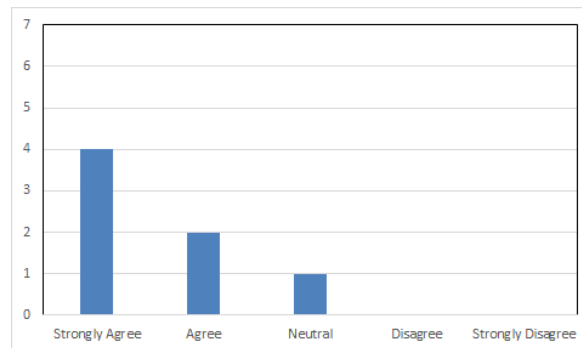
The laboratory program was useful in supporting, reinforcing, or augmenting my understanding of course content:



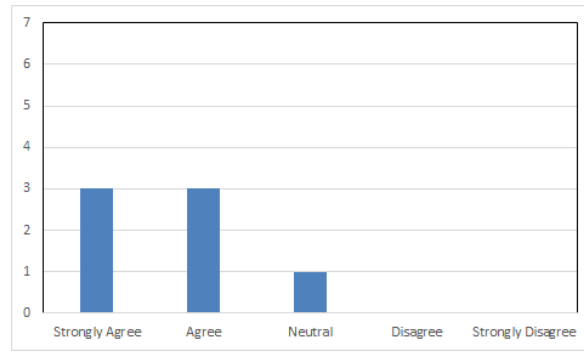
The course text and references were very helpful in my gaining understanding of the course material:



The Course Goals and Lesson Objectives for this course gave me a clear understanding of what was expected of me in order to do well in this course:



The WPRs, block exams, and quizzes in this course were a reasonable evaluation of my understanding of the course material:



SECTION II – ATTACHMENT 2

HANDWRITTEN SURVEY YEARLY COMPARISON

SECTION II – ATTACHMENT 3
STUDENT OUTCOMES WORKSHEETS

	A	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T
		PS1	PS2	PS3	PS4	PS5	PS		Capstone Present	Capstone Paper	Project		Quiz1	Quiz2	Quiz3	Quiz4	Quiz5	Quiz
1	Coverage																	
2	Student Outcomes																	
3	(Outcomes are abbreviated)	← Categories Carried Forward from Grades Spreadsheet																
4																		
5	On completion of the program, our students have demonstrated that they can:																	
6																		
7	id, form & solve by appl mse	0	0	1	1	1	0.6		0	0	0.0		0	0	1	1	0	0.4
8	design to meet needs	1	0	0	1	0	0.4		0	0	0.0		0	0	0	0	0	0.0
9	communicate	0	0	0	0	0	0.0		1	1	1.0		0	0	0	0	0	0.0
10	professional ethical impact	0	0	0	0	0	0.0		0	0	0.0		0	0	0	0	0	0.0
11	function on a team	0	0	0	0	0	0.0		0	0	0.0		0	0	0	0	0	0.0
12	experiments	0	0	0	0	0	0.0		0	0	0.0		0	0	0	0	0	0.0
13	new knowledge	1	1	1	1	1	1.0		1	1	1.0		0	0	0	0	0	0.0
14																		
15	The program provides a thorough grounding and working knowledge of:																	
16																		
17	chemistry	0	1	1	1	1	0.8		1	1	1.0		0	1	1	1	1	0.8
18	material & energy balances	0	0	0	1	0	0.2		1	1	1.0		0	0	0	1	0	0.2
19	safety & environmental	0	0	0	0	0	0.0		0	0	0.0		0	0	0	0	0	0.0
20	phys. & chem. thermo	1	0	0	0	0	0.2		0	0	0.0		Chart Area		0	0	0	0.2
21	heat, mass, & momentum transfer.	0	0	1	0	0	0.2		0	0	0.0		0	0	1	0	0	0.2
22	chemical reaction engineering	0	0	1	1	0	0.4		0	0	0.0		0	0	1	1	0	0.4
23	cont. & stage-wise separations	0	0	0	0	0	0.0		0	0	0.0		0	0	0	0	0	0.0
24																		



UNITED STATES MILITARY ACADEMY
WEST POINT.

SECTION II

BLOCK QUIZZES



TAB C

BLOCK QUIZ #1

Handouts

**“Gait Analysis”
Biomechanics Block Quiz**

A Problem Worksheet Submitted to:

Instructor/Section: Dr. Yuk/

In Partial Fulfillment of

Requirements for CH450

By

Cadet: _____

West Point, New York

Date: _____

You are not authorized to discuss or communicate about the exam in any manner (verbal, written, etc.) until the Quiz has been released from academic security, which will occur when the quiz is returned to all the cadets. You may use any references **other than another cadet and internet.** Ensure that you parenthetically cite references, including the course textbook. **No point will be given** for any late submission. **Sign below to acknowledge:**

SIGNATURES: _____

a) **(15 pts)** Pelvic obliquity is commonly associated with an angular displacement of the pelvis as seen from the front of the male patient and is estimated as the angle formed between the vector from right to left anterior superior iliac spine (ASIS) and the horizon. Given the pelvic position data below, compute **an anatomical coordinate system** for the pelvis.

$$\text{Right ASIS} = 0.400 \mathbf{i} + 0.400 \mathbf{j} + 0.580 \mathbf{k}$$

$$\text{Left ASIS} = 0.390 \mathbf{i} + 0.435 \mathbf{j} + 0.600 \mathbf{k}$$

$$\text{PSIS} = 0.100 \mathbf{i} + 0.418 \mathbf{j} + 0.820 \mathbf{k}$$

Provide either the hand-written or Mathematica solution as part of the submission.

b) **(15 pts)** Now, a female patient performs 30 degrees of internal transverse rotation, then 20 degrees of extension, and finally 10 degrees of abduction during a subsequent observation session. Walking produces tri-planar hip, knee, and ankle motion: flexion/extension (θ_y), abduction/adduction (θ_x), and internal/external transverse rotation (θ_z). You are tasked to provide the **Euler angle transformation matrix** for this series of angular rotations to the clinician. Again, provide either the hand-written or Mathematica solution as part of the submission.

Instructor Solution

**“Gait Analysis”
Biomechanics Block Quiz**

A Problem Worksheet Submitted to:

Instructor/Section: Dr. Yuk/

In Partial Fulfillment of

Requirements for CH450

By

Cadet: APPROVED SOLUTION

West Point, New York

Date: _____

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SIGNATURES: _____

a) **(15 pts)** Pelvic obliquity is commonly associated with an angular displacement of the pelvis as seen from the front of the male patient and is estimated as the angle formed between the vector from right to left anterior superior iliac spine (ASIS) and the horizon. Given the pelvic position data below, compute **an anatomical coordinate system** for the pelvis.

$$\begin{aligned}\text{Right ASIS} &= 0.400 \mathbf{i} + 0.400 \mathbf{j} + 0.580 \mathbf{k} \\ \text{Left ASIS} &= 0.390 \mathbf{i} + 0.435 \mathbf{j} + 0.600 \mathbf{k} \\ \text{PSIS} &= 0.100 \mathbf{i} + 0.418 \mathbf{j} + 0.820 \mathbf{k}\end{aligned}$$

Provide either the hand-written or Mathematica solution as part of the submission. **(Refer to Ch. 4.2 (p. 138, 146-148) and Ch. 4.6 (p. 177-179))**

```
In[1]:= (*CH450 Quiz 1 part a Soln*)
```

```
In[2]:= RASIS = {0.400, 0.400, 0.580};
```

```
LASIS = {0.390, 0.435, 0.600};
```

```
PSIS = {0.100, 0.418, 0.820};
```

```
r1 = LASIS - RASIS;
```

```
epay = r1 / Norm[r1];
```

+2 pts each for correctly calculating the x, y, z vector; 6 pts total.

```
r2 = PSIS - RASIS;
```

+1 pts for correctly calculating the x, y, z unit vector; 6 pts total.

```
r3 = Cross[r1, r2];
```

+3 pts for the final matrix showing correct anatomical coordinate system based on unit vector

```
epaz = r3 / Norm[r3];
```

```
epax = Cross[epay, epaz];
```

```
epa = {epax, epay, epaz} // MatrixForm
```

```
Out[11]//MatrixForm=
```

$$\begin{pmatrix} 0.76871 & 0.468466 & -0.43546 \\ -0.240772 & 0.842701 & 0.481543 \\ 0.592549 & -0.265321 & 0.760586 \end{pmatrix}$$

b) **(15 pts)** Now, a female patient performs 30 degrees of internal transverse rotation, then 20 degrees of extension, and finally 10 degrees of abduction during a subsequent observation session. Walking produces tri-planar hip, knee, and ankle motion: flexion/extension (θ_y), abduction/adduction (θ_x), and internal/external transverse rotation (θ_z). You are tasked to provide the **Euler angle transformation matrix** for this series of angular rotations to the clinician. Again, provide either the hand-written or Mathematica solution as part of the submission. **(Refer to Ch. 4.2 (p. 138, 146-148))**

```
In[1]:= (*CH450 Quiz 1 part b Soln*)
(*Given Angles of Rotation*)
 $\theta_x = N[\pi / 18, 10];$ 
 $\theta_y = N[\pi / 9, 10];$ 
 $\theta_z = N[\pi / 6, 10];$ 

(*x, y, and z-axis Transformation Matrix*)
XTrans = {{1, 0, 0}, {0, Cos[ $\theta_x$ ], Sin[ $\theta_x$ ]}, {0, -Sin[ $\theta_x$ ], Cos[ $\theta_x$ ]}};
YTrans = {{Cos[ $\theta_y$ ], 0, -Sin[ $\theta_y$ ]}, {0, 1, 0}, {Sin[ $\theta_y$ ], 0, Cos[ $\theta_y$ ]}};
ZTrans = {{Cos[ $\theta_z$ ], Sin[ $\theta_z$ ], 0}, {-Sin[ $\theta_z$ ], Cos[ $\theta_z$ ], 0}, {0, 0, 1}};

(*Solution :: z-y-x axis rotation*)
DTrans = XTrans.YTrans.ZTrans // MatrixForm

Out[7]//MatrixForm=
```

$$\begin{pmatrix} 0.8137976813 & 0.4698463104 & -0.3420201433 \\ -0.4409696105 & 0.8825641193 & 0.1631759112 \\ 0.3785223064 & 0.0180283112 & 0.9254165784 \end{pmatrix}$$

+1 pts each for each x, y, and z angle input (in radian or degree); 3 pts total.

+2 pts each for x, y, and z transformation matrix; 6 pts total.

+3 pts for correctly defining the order of matrix operation.

+3 pts for numerical accuracy of final answer.

-5 pts for simply using a Mathematica built-in EulerMatrix symbol (definition of transformation matrix in Mathematica)



TAB D BLOCK QUIZ #2

Handouts

**“Biomaterials for Human Implantations”
Biomaterials Block Quiz**

A Problem Worksheet Submitted to:

Instructor/Section: Dr. Yuk/

In Partial Fulfillment of

Requirements for CH450

By

Cadet: _____

West Point, New York

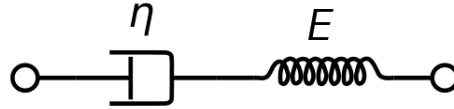
Date: _____

You are not authorized to discuss or communicate about the exam in any manner (verbal, written, etc.) until the Quiz has been released from academic security, which will occur when the quiz is returned to all the cadets. You may use any references **other than another cadet and internet.** Ensure that you parenthetically cite references, including the course textbook. **No point will be given** for any late submission. **Sign below to acknowledge:**

SIGNATURES: _____

a) **(10 pts)** With interest in the biological self-assembly of biopolymers, organelles, cells, tissues, and organs, as bioengineering tools, explain why after shaking vinaigrette dressing, oil droplets coalesce back into the water layers. Use intermolecular forces, entropy, and Gibbs free energy concepts to explain. What governs the rate at which this process occurs?

b) **(20 pts)** As an entry-level bioengineer intern, you are tasked to analyze a viscoelastic tissue using the Maxwell model (a linear spring with modulus E and a dashpot with viscosity η connected in series). Your goal is to derive the equation for stress relaxation $\sigma(t)$ when a constant strain ϵ_0 is applied at $t = 0$. Your final answer should be in the form of an exponential decay equation, $\sigma(t) = \sigma_0 e^{-t/\tau}$.



Instructor Solution

**“Biomaterials for Human Implantations”
Biomaterials Block Quiz**

A Problem Worksheet Submitted to:

Instructor/Section: Dr. Yuk/

In Partial Fulfillment of

Requirements for CH450

By

Cadet: APPROVED SOLUTION

West Point, New York

Date: _____

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SIGNATURES: _____

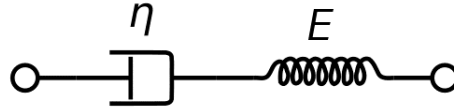
a) **(10 pts)** With interest in the biological self-assembly of biopolymers, organelles, cells, tissues, and organs, as bioengineering tools, explain why after shaking vinaigrette dressing, oil droplets coalesce back into the water layers. Use intermolecular forces, entropy, and Gibbs free energy concepts to explain. What governs the rate at which this process occurs?

Answers can be varied, but the following aspects must be discussed to obtain full points:

- 1) $\Delta G = \Delta H - T\Delta S$; the competition between enthalpic and entropic contribution needs to be discussed. If ΔH is assumed 0, the reasoning behind such assumption needs to be written.**
- 2) Intermolecular forces of H₂O and oil must be mentioned; hydrogen bonding is the key difference in IMF between H₂O and oil. Lewis structures can be provided to further discuss the polarity of these molecules.**
- 3) Since oil will disrupt the hydrogen bonding between H₂O, H₂O will form a “cage” around oil to minimize the surface area interacting with oil and thus maximize ΔS in the process. This aspect must be mentioned to discuss why oil droplets coalesce in the H₂O layer, instead of spreading apart.**
- 4) Diffusion rate of oil droplets coalescing in H₂O layer will be impacted by multiple factors: temperature, the electrostatic interaction between H₂O and oil, etc.**

+4 pts for discussing about the competition between enthalpic and entropic contribution.
+3 pts for discussing about the IMF between H₂O and oil.
+3 pts for describing the entropic contribution in detail.

b) **(20 pts)** As an entry-level bioengineer intern, you are tasked to analyze a viscoelastic tissue using the Maxwell model (a linear spring with modulus E and a dashpot with viscosity η connected in series). Your goal is to derive the equation for stress relaxation $\sigma(t)$ when a constant strain ϵ_0 is applied at $t = 0$. Your final answer should be in the form of an exponential decay equation, $\sigma(t) = \sigma_0 e^{-t/\tau}$. **(Refer to pg. 8 to 10 of Viscoelasticity, Roylance, 2001 for full information)**



The Maxwell model consists of a linear spring (elastic) and a linear dashpot (viscous) connected in series. Because they are in series:

$$\sigma = \sigma_{spring} = \sigma_{dashpot}$$

$$\epsilon = \epsilon_{spring} + \epsilon_{dashpot}$$

To find the equation for the whole system, we look at the rates of change. So, we differentiate the strain equation by taking the time derivative of the additive strain rule:

$$\dot{\epsilon} = \dot{\epsilon}_{spring} + \dot{\epsilon}_{dashpot}$$

$$\epsilon_{spring} = \frac{\sigma}{E}; \quad \dot{\epsilon}_{spring} = \frac{\dot{\sigma}}{E} \text{ (Hooke's Law)}$$

$$\sigma = \eta \dot{\epsilon}_{dashpot}; \quad \dot{\epsilon}_{dashpot} = \frac{\dot{\sigma}}{\eta} \text{ (Newtonian Fluid)}$$

Then,

$$\dot{\epsilon} = \frac{\dot{\sigma}}{E} + \frac{\sigma}{\eta}$$

$$E\dot{\epsilon} = \dot{\sigma} + \frac{\sigma}{\tau} \text{ (with } \tau = \frac{\eta}{E}\text{)}$$

Since $\dot{\epsilon} = 0$ during stress relaxation, you will get

$$0 = \dot{\sigma} + \frac{\sigma}{\tau}$$

$$\frac{d\sigma}{dt} = -\frac{\sigma}{\tau}$$

$$\int_{\sigma_0}^{\sigma(t)} \frac{d\sigma}{\sigma} = -\frac{1}{\tau} \int_0^t dt$$

$$\ln \sigma(t) - \ln \sigma_0 = -\frac{t}{\tau}$$

$$\sigma(t) = \sigma_0 e^{-t/\tau}$$

+5 pts for correctly laying out the spring vs. dashpot equations.

+5 pts for correctly differentiating the strain equations.

+5 pts for correctly summarizing the strain vs. time relationship.

+5 pts for correctly deriving the final equation shown in the problem statement.



TAB E BLOCK QUIZ #3

Handouts

**“Cell Differentiation”
Tissue Engineering Block Quiz**

A Problem Worksheet Submitted to:

Instructor/Section: Dr. Yuk/

In Partial Fulfillment of

Requirements for CH450

By

Cadet: _____

West Point, New York

Date: _____

You are not authorized to discuss or communicate about the exam in any manner (verbal, written, etc.) until the Quiz has been released from academic security, which will occur when the quiz is returned to all the cadets. You may use any references **other than another cadet and internet.** Ensure that you parenthetically cite references, including the course textbook. **No point will be given** for any late submission. **Sign below to acknowledge:**

SIGNATURES: _____

a) **(10 pts)** Describe the various stages in cell differentiation. Describe the general signaling process that leads to cell differentiation.

b) **(20 pts)** Mathematically describe the cell population as a function of time of the following notional cell lineage. State any assumption if needed:

$$A \leftrightarrow B \leftrightarrow C$$

Instructor Solution

**“Cell Differentiation”
Tissue Engineering Block Quiz**

A Problem Worksheet Submitted to:

Instructor/Section: Dr. Yuk/

**In Partial Fulfillment of
Requirements for CH450**

By

Cadet: _____

West Point, New York

Date: _____

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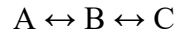
SIGNATURES: _____

a) **(10 pts)** Describe the various stages in cell differentiation. Describe the general signaling process that leads to cell differentiation.

The answer can be varied, but the following key points need to be discussed:

Cell differentiation is the method by which a less specialized cell becomes a more specialized cell type. It doesn't happen all at once; the differentiation occurs through a series of progressively restrictive "potency" stages, often visualized as a ball rolling down a valley. Cell differentiation is triggered by extracellular signals that tell the cell which genes to turn on or off. This process typically follows a four-step signaling cascade: signal reception, transduction, transcription factor activation, and epigenetic modification/gene expression.

b) **(20 pts)** Mathematically describe the cell population as a function of time of the following notional cell lineage. State any assumption if needed:



$$\frac{d[A]}{dt} = -k_1[A] + k_2[B]$$

$$\frac{d[B]}{dt} = k_1[A] - (k_2 + k_3)[B] + k_4[C]$$

$$\frac{d[C]}{dt} = k_3[B] - k_4[C]$$

The model assumes that the rate of transition between states is linearly proportional to the population size of the source state (elementary step). Also, it is assumed that the cells are in a "well-mixed" environment. The model additionally assumes a close system, meaning there is no cell division (proliferation) and death (apoptosis). Lastly, the rate constants are assumed to be time and temperature independent.



TAB F BLOCK QUIZ #4

Handouts

“Enzyme Inhibition Kinetics”
Biochemical Reactions and Enzyme Kinetics Block Quiz

A Problem Worksheet Submitted to:

Instructor/Section: Dr. Yuk/

In Partial Fulfillment of

Requirements for CH450

By

Cadet: _____

West Point, New York

Date: _____

You are not authorized to discuss or communicate about the exam in any manner (verbal, written, etc.) until the Quiz has been released from academic security, which will occur when the quiz is returned to all the cadets. You may use any references **other than another cadet and internet.** Ensure that you parenthetically cite references, including the course textbook. **No point will be given** for any late submission. **Sign below to acknowledge:**

SIGNATURES: _____

(30 pts) Develop batch design equations of the following enzyme inhibition kinetics. **Plot the concentration profiles** of E, S, I, ES, EI, ESI, and P as a function of time (set $t_{\max} = 10$ min).

$$k_1 = 1/(\text{M} \cdot \text{min})$$

$$k_2 = 0.5/\text{min}$$

$$k_3 = 1/(\text{M} \cdot \text{min})$$

$$k_4 = 0.5/\text{min}$$

$$k_5 = 1/\text{min}$$

$$k_6 = 1/(\text{M} \cdot \text{min})$$

$$k_7 = 1/\text{min}$$

$$k_8 = 1/(\text{M} \cdot \text{min})$$

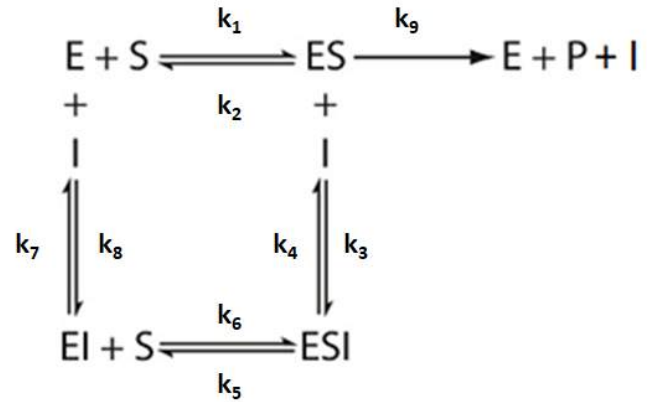
$$k_9 = 3/\text{min}$$

At $t = 0$,

$$C_E = C_S = 1.0 \text{ M}$$

$$C_I = 0.5 \text{ M}$$

$$C_P = C_{ES} = C_{EI} = C_{ESI} = 0.0 \text{ M}$$



Instructor Solution

“Enzyme Inhibition Kinetics”
Biochemical Reactions and Enzyme Kinetics Block Quiz

A Problem Worksheet Submitted to:

Instructor/Section: Dr. Yuk/

In Partial Fulfillment of

Requirements for CH450

By

Cadet: APPROVED SOLUTION

West Point, New York

Date: _____

You are not authorized to discuss or communicate about the exam in any manner (verbal, written, etc.) until the Quiz has been released from academic security, which will occur when the quiz is returned to all the cadets. You may use any references other than another cadet and internet. Ensure that you parenthetically cite references, including the course textbook. No point will be given for any late submission. Sign below to acknowledge:

SIGNATURES: _____

(30 pts) Develop batch design equations of the following enzyme inhibition kinetics. **Plot the concentration profiles** of E, S, I, ES, EI, ESI, and P as a function of time (set $t_{\max} = 10$ min).

$$k_1 = 1/(\text{M} \cdot \text{min})$$

$$k_2 = 0.5/\text{min}$$

$$k_3 = 1/(\text{M} \cdot \text{min})$$

$$k_4 = 0.5/\text{min}$$

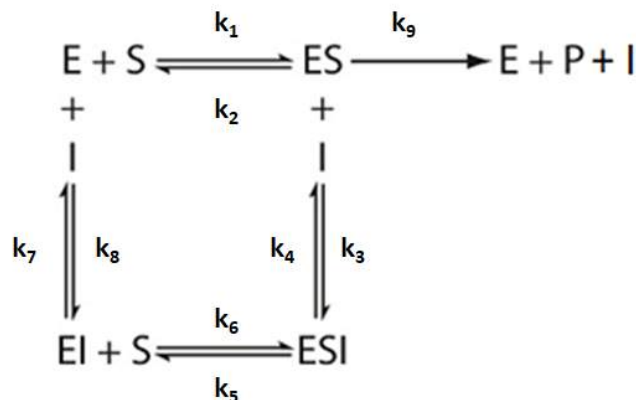
$$k_5 = 1/\text{min}$$

$$k_6 = 1/(\text{M} \cdot \text{min})$$

$$k_7 = 1/\text{min}$$

$$k_8 = 1/(\text{M} \cdot \text{min})$$

$$k_9 = 3/\text{min}$$



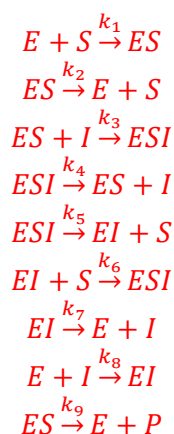
At $t = 0$,

$$C_E = C_S = 1.0 \text{ M}$$

$$C_I = 0.5 \text{ M}$$

$$C_P = C_{ES} = C_{EI} = C_{ESI} = 0.0 \text{ M}$$

The above reaction mechanism can be broken into individual steps:



Then,

$$\frac{dC_E}{dt} = -k_1 C_E C_S + k_2 C_{ES} + k_7 C_{EI} - k_8 C_E C_I + k_9 C_{ES}$$

$$\frac{dC_S}{dt} = -k_1 C_E C_S + k_2 C_{ES} + k_5 C_{ESI} - k_6 C_{EI} C_S$$

$$\frac{dC_P}{dt} = k_9 C_{ES}$$

$$\frac{dC_I}{dt} = -k_3 C_{ES} C_I + k_4 C_{ESI} + k_7 C_{EI} - k_8 C_E C_I$$

$$\frac{dC_{ES}}{dt} = k_1 C_E C_S - k_2 C_{ES} - k_3 C_{ES} C_I + k_4 C_{ESI} - k_9 C_{ES}$$

$$\frac{dC_{EI}}{dt} = k_5 C_{ESI} - k_6 C_{EI} C_S - k_7 C_{EI} + k_8 C_E C_I$$

$$\frac{dC_{ESI}}{dt} = k_3 C_{ES} C_I - k_4 C_{ESI} - k_5 C_{ESI} + k_6 C_{EI} C_S$$

AY26-2

“Enzyme Inhibition Kinetics” Quiz

Due: NLT 11:50, 24 MAR 2026

In[]:= (*CH450 Enzyme Kinetics Inhibition Quiz*)

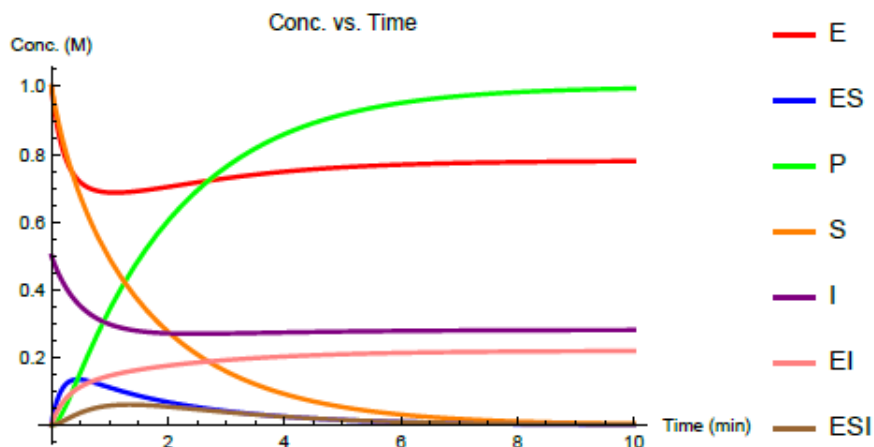
```

k1 = 1; (*1/(M*min)*)
k2 = 0.5; (*1/min*)
k3 = 1; (*1/(M*min)*)
k4 = 0.5; (*1/min*)
k5 = 1; (*1/min*)
k6 = 1; (*1/(M*min)*)
k7 = 1; (*1/min*)
k8 = 1; (*1/(M*min)*)
k9 = 3; (*1/min*)
sol = NDSolve[
  {CE'[t] == -k1*CE[t]*CS[t] + k2*CES[t] + k7*CEI[t] - k8*CE[t]*CI[t] + k9*CES[t],
   CS'[t] == -k1*CE[t]*CS[t] + k2*CES[t] + k5*CESI[t] - k6*CEI[t]*CS[t], CP'[t] ==
    k9*CES[t], CI'[t] == -k3*CES[t]*CI[t] + k4*CESI[t] + k7*CEI[t] - k8*CE[t]*CI[t],
   CES'[t] == k1*CE[t]*CS[t] - k2*CES[t] - k3*CES[t]*CI[t] + k4*CESI[t] - k9*CES[t],
   CEI'[t] == k5*CESI[t] - k6*CEI[t]*CS[t] - k7*CEI[t] + k8*CE[t]*CI[t],
   CESI'[t] == k3*CES[t]*CI[t] - k4*CESI[t] - k5*CESI[t] + k6*CEI[t]*CS[t],
   CE[0] == 1, CS[0] == 1, CES[0] == 0, CP[0] == 0, CI[0] == 0.5, CEI[0] == 0, CESI[0] == 0},
  {CE[t], CP[t], CS[t], CI[t], CES[t], CEI[t], CESI[t]}, {t, 0, 10}];
F[t_] = CE[t] /. sol;
G[t_] = CES[t] /. sol;
H[t_] = CP[t] /. sol;
J[t_] = CS[t] /. sol;
A[t_] = CI[t] /. sol;
B[t_] = CEI[t] /. sol;
L[t_] = CESI[t] /. sol;

p1 = Plot[F[t], {t, 0, 10}, PlotRange -> Full, PlotStyle -> Red, PlotLegends -> {"E"}];
p2 = Plot[G[t], {t, 0, 10}, PlotRange -> Full, PlotStyle -> Blue, PlotLegends -> {"ES"}];
p3 = Plot[H[t], {t, 0, 10}, PlotRange -> Full, PlotStyle -> Green, PlotLegends -> {"P"}];
p4 = Plot[J[t], {t, 0, 10}, PlotRange -> Full, PlotStyle -> Orange, PlotLegends -> {"S"}];
p5 = Plot[A[t], {t, 0, 10}, PlotRange -> Full, PlotStyle -> Purple, PlotLegends -> {"I"}];
p6 = Plot[B[t], {t, 0, 10}, PlotRange -> Full, PlotStyle -> Pink, PlotLegends -> {"EI"}];
p7 = Plot[L[t], {t, 0, 10}, PlotRange -> Full, PlotStyle -> Brown, PlotLegends -> {"ESI"}];
Show[p1, p2, p3, p4, p5, p6, p7, PlotRange -> {{0, 10}, {0, 1}},
  AxesLabel -> {"Time (min)", "Conc. (M)"},
  PlotLabel -> "Conc. vs. Time", AxesOrigin -> {0, 0}]

```

Out[]:=





TAB G

BLOCK QUIZ #5

Handouts

“Biomedical Sensors”
Biomedical Sensors and Bio-Imaging Block Quiz

A Problem Worksheet Submitted to:

Instructor/Section: Dr. Yuk/

In Partial Fulfillment of

Requirements for CH450

By

Cadet: _____

West Point, New York

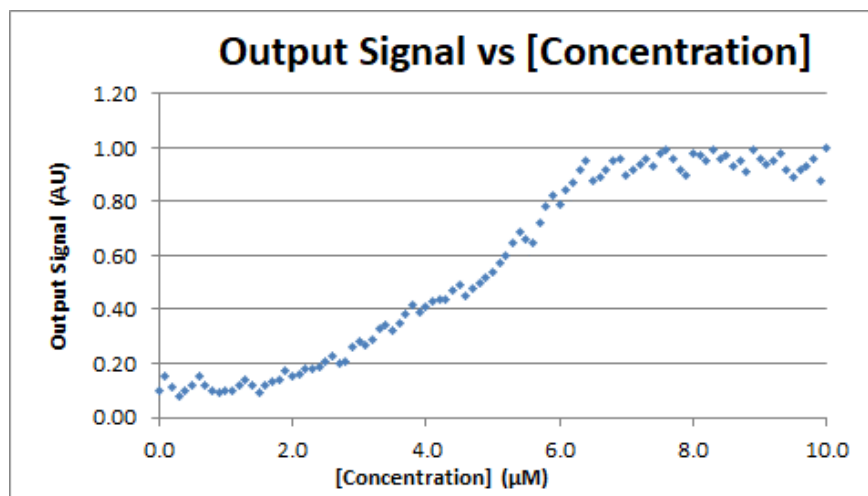
Date: _____

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SIGNATURES: _____

(20 pts) For the Output Signal (Arbitrary Units) versus Concentration (μM) data of typical biomedical sensor in the plot shown below, answer the following questions:

- Describe the mathematical form of the data.
- Mathematically describe the sensitivity for this data.
- What are possible reasons for the data not intersecting the origin?
- What does the data region from concentrations 0.0-2.0 μM indicate?
- What sort of error (s) and/or phenomenon result in the data in the region with concentrations 6.0-10.0 μM ?
- Describe the precision and accuracy of the data.



(10 pts) What does life-long learning mean to you? How well has your West Point experience prepared you for life-long learning?

Instructor Solution

“Biomedical Sensors”
Biomedical Sensors and Bio-Imaging Block Quiz

A Problem Worksheet Submitted to:

Instructor/Section: Dr. Yuk/

In Partial Fulfillment of

Requirements for CH450

By

Cadet: APPROVED SOLUTION

West Point, New York

Date: _____

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SIGNATURES: _____

(20 pts) For the Output Signal (Arbitrary Units) versus Concentration (μM) data of typical biomedical sensor in the plot shown below, answer the following questions:

a) Describe the mathematical form of the data.

The data exhibits a piecewise functional form, consisting of a combination of linear and nonlinear trends. Specifically, it shows a low-gradient initial phase, a linear high-sensitivity region, and an eventual nonlinear plateau (asymptotic behavior).

b) Mathematically describe the sensitivity for this data.

Sensitivity is defined as the change in output signal per unit change in concentration, represented by the slope of the curve. Because the relationship is not uniform, sensitivity must be calculated locally for each distinct region of the data set to account for changing gradients

c) What are possible reasons for the data not intersecting the origin?

A non-zero intercept ($y\text{-intercept} > 0$) indicates a baseline offset or background signal. Possible causes include improperly calibrated equipment and interference from ambient temperature, pressure, or electronic noise.

d) What does the data region from concentrations 0.0-2.0 μM indicate?

This is a low-sensitivity region, likely representing the sensor's threshold or approach to the limit of detection.

e) What sort of error (s) and/or phenomenon result in the data in the region with concentrations 6.0-10.0 μM ?

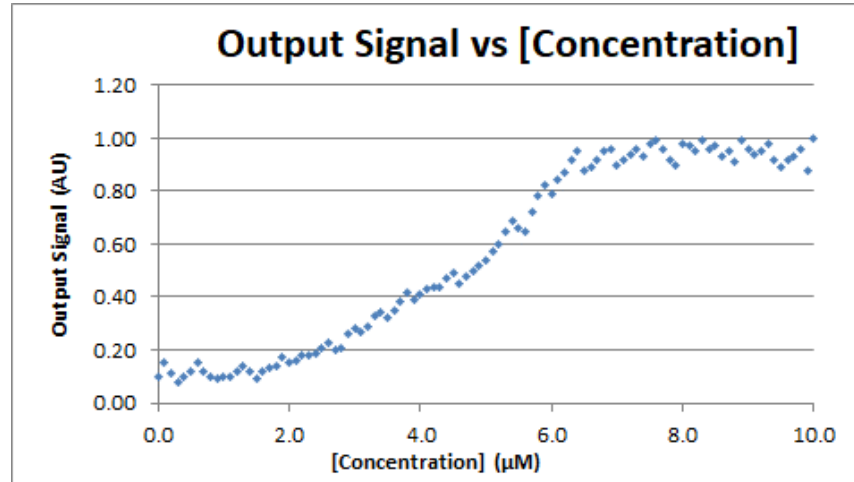
The output signal becomes independent of concentration (slope approaches zero). Most likely, the detector has reached its maximum physical capacity to respond, or all available binding sites on the sensor are occupied.

f) Describe the precision and accuracy of the data.

Answers can be varied, but must be tied directly to the provided data points rather than abstract definitions:

Precision: Assessed by the reproducibility of the data points (look for the spread of error bars or "tightness" of repeated trials).

Accuracy: Assessed by how closely the data aligns with a known reference standard or theoretical model.



(10 pts) What does life-long learning mean to you? How well has your West Point experience prepared you for life-long learning?

Answers can be varied, but cadet needs to clearly define what the meaning and importance of life-long learning is for them in their army career.



TAB H **FINAL COURSE GRADES**

Cadet	Major	Minor	IP1	IP2	IP4	IP4	IP5	PS1	PS2	PS3	PS4	PS5	Quiz 1	Quiz 2	Quiz 3	Quiz 4	Quiz 5	CP IPR	CP Pres	CP Pap	Current Points	Percentage	Letter Grade	OML
Delia, James	BIO		20	20	20	20	20	98	100	100	100	100	30	30	26	29	26	45	47	144	975	97.5	A+	1
Davis, Ella	BIO		20	20	20	20	20	100	95	97	100	98	28	25	26	26.5	30	50	50	148	973.5	97.35	A+	2
Davis, Nathan	CEN		20	20	20	20	20	100	98	100	100	98	28	24	28	27	27	50	50	140	970	97	A+	3
Littlehales, Andrew	CEN		20	20	20	20	19	97	95	90	97	92	28	30	30	29.5	29	50	50	145	961.5	96.15	A	4
Badger, James	CEN		20	20	20	20	19	95	86	97	97	94	26	30	30	30	30	50	50	145	959	95.9	A	5
Barnes, Alexis	BIO	RSE	20	19	20	19	19	94	97	97	100	100	28	27	26	28	28	45	45	146	958	95.8	A	6
Feng, Xinyi	BIO		20	20	19	19	19	95	96	95	97	92	28	28	26	27	26	50	50	148	955	95.5	A	7
Mantooth, Jeanine	CEN		20	20	20	20	20	97	90	91	97	100	28	29	27	24	28	45	45	146	947	94.7	A	8
Chapman, Wyatt	BIO		20	19	19	20	20	98	82	87	100	98	28	26	26	27	28	45	47	144	933.5	93.35	A	9
Macune, Grace	CEN		20	19	19	20	20	96	84	87	97	87	28	27	26	29.5	27	50	50	140	926.5	92.65	A-	10
Janat, Tsegaye	CEN		20	20	20	19	19	93	94	80	94	87	28	28	29.5	29.5	29	45	46	140	921	92.1	A-	11
Siomades, Alexander	EVE	BEN	20	20	18	19	19	87	97	88	92	83	28	30	30	29.5	28	45	49	130	912.5	84.83	A-	12
Ferrying, Zane	CEN		19	20	18	19	18	94	94	87	94	87	28	28	26	29.5	26	40	45	136	908.5	90.85	A-	13
Baisted, Benjamin	CEN		19	19	18	19	20	95	95	72	97	92	28	28	25	26.5	29	40	45	136	903.5	90.35	A-	14
Marcellus, Calvin	BIO		20	19	19	18	20	95	84	83	94	82	26	19	23	25.5	23	50	42	127	869.5	86.95	B	15
Ebiuwhe, Emmanuella	CEN		20	19	20	19	18	94	70	87	82	70	28	25	25	27	28	45	46	140	863	86.3	B	16
Orfao, Daniel	BIO		20	19	19	18	19	95	80	84	78	87	26	24	25	18	27	50	42	127	858	85.8	B	17
Leja, Gabriel	CEN		19	20	18	19	18	93	70	69	87	73	28	24.5	26	29	29	40	45	136	843.5	84.35	B	18



UNITED STATES MILITARY ACADEMY
WEST POINT.

SECTION III

ASSIGNMENTS



TAB I PROBLEM SETS



PROBLEM SET #1

Handouts

PROBLEM SET 1: Biomechanics**Knee Brace Energy Harvester****References:**

The following references are intended to serve as a starting point to analyze the questions below. You may use any additional references and you are encouraged to collaborate with your classmates. Ensure that all referenced material and collaboration is properly documented.

1. Introduction to Biomedical Engineering, Sections 4.6. Clinical Gait Analysis.
2. Biomechanical Energy Harvesting: Generating Electricity During Walking with Minimal User Effort, Donelan (2008).
3. Biomechanical Energy Harvesting: Apparatus and Method, Li (2008).
4. Optimal design of gearbox for application in knee mounted biomechanical energy harvester, Jhalani (2012).

Canvas website link for references:

<https://westpoint.instructure.com/courses/11109>

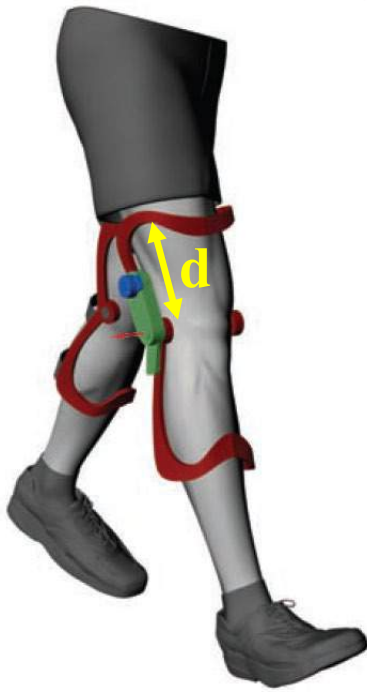
Background:

The Army is funding research of energy harvesting devices, those devices that convert “waste” energy to usable electricity. One candidate is the knee brace energy harvester. The brace affixes to the knee joint in a similar manner to an orthotic brace and is able to convert the “negative work” associated with the eccentric motion of the leg to electricity via an electromagnetic generator.

The purpose of this problem set is to integrate the concepts of the biomechanics block to analyze various aspects of the device to include human-device interface, human-external energy system relationships, and practical implementation. When applicable, answers should include a mathematical model and quantitative analysis.

Questions:

1. Discuss practical and technical issues with implementation of a knee harvesting device for use by U.S. Army Soldiers in tactical combat scenarios for both real-time powering of devices and recharging of rechargeable batteries. Answer should be no more than 2 pages long.
2. Given the concepts of torque and pressure, what is the practical consequence with the length away from the knee joint (d) that the knee brace energy harvester contacts the leg (see figure below)? Model the pressure of the aluminum attachment cross braces on the leg as a function of distance from the knee joint, and brace width and construct a 3-dimensional X-Y-Z plot of Pressure versus Knee Joint Distance versus Brace Width. What are the design trade-offs due to widening the attachment cross braces?



Biomechanical Energy Harvesting, Donelan (2008)

3. Given your results from Problem 2 using the prototype aluminum frame knee brace, what materials would you use to redesign the knee brace with? Consider different strategies to address possible issues identified in Problem 1. For the selected material, and optimized cross-brace distance from the knee and brace width, what is the predicted material deformation?
4. Given the energy conversion estimates offered in the literature, identify any assumptions, explicit or implicit, that may not be valid and will result in additional energy losses. **Model and provide an estimate for the most significant additional loss.**
5. (*Optional-Extra Credit*). Given the generator system used in the referenced articles, what is the effect of increasing the gear ratio on the conversion of mechanical to electrical work? Model the increase of gear ratio with the predicted increase in mass and decrease in energy conversion. A 3-dimensional plot might be appropriate to show these relationships.

Guidance:

1. Clearly state all assumptions for all answers.
2. Include derivations for all relationships that are modeled.
3. Clearly label and annotate figures, tables, and schematics.
4. Derivations and schematics may be hand written as long as they are neatly presented.

Report Format:

1. References. Use the CSE citation-sequence. See the following site as a guide, <http://www.wsulibs.wsu.edu/quickguides/cse>.

Instructor Solution

NOT PROVIDED



PROBLEM SET #2

Handouts

PROBLEM SET 2: Biomaterials**Viscoelastic Modeling of Cartilage****References:**

The following references are intended to serve as a starting point to analyze the questions below. You may use any additional references and you are encouraged to collaborate with your classmates. Ensure that all referenced material and collaboration is properly documented.

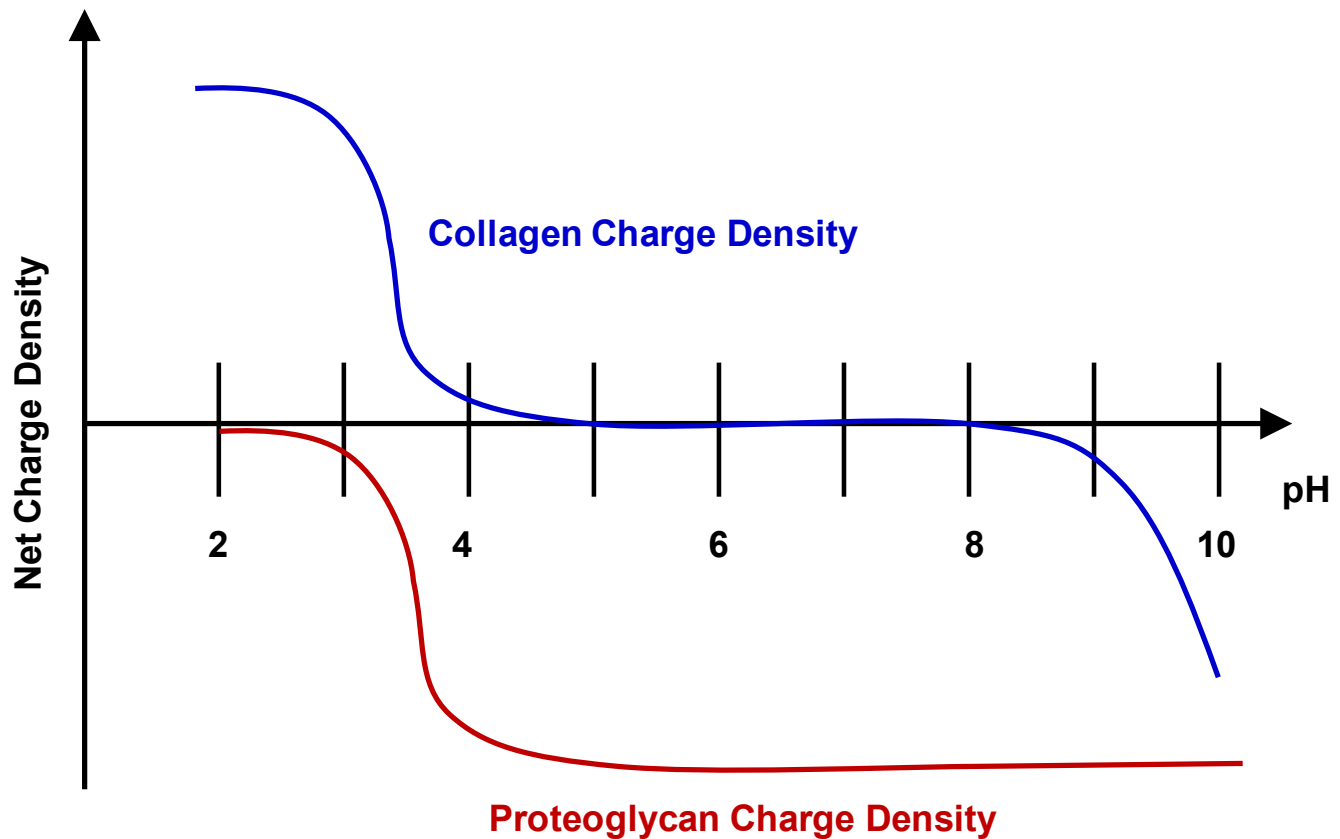
1. Introduction to Biomedical Engineering, Sections 4.4. Viscoelastic Properties.
2. Engineering Viscoelasticity, Roylance (2001). The pdf version of scanned file is located in <https://westpoint.instructure.com/courses/11109/pages/course-resources>.
3. Biomechanical Behavior of the Temporomandibular Joint Disc, Tanaka (2003).
4. Biomechanics of Articular Cartilage and Determination of Material Properties, Lu (2008).
5. Nanocomposite Polymer Biomaterials for Tissue Repair of Bone and Cartilage: A Material Science Perspective, Gaharwar.
6. Viscoelastic properties of bovine articular cartilage attached to subchondral bone at high frequencies, Fulcher (2009).

Background:

Progressive cartilage damage and disease is a ubiquitous problem for Soldiers and civilians alike. To develop novel strategies for artificial cartilage replacements, an understanding of viscoelastic materials is necessary. As a starting point, simplified Maxwell, Voight, and Kelvin models will be explored to model the response of candidate materials. Modeling biomaterial behavior to loading will serve as a bridge between the biomechanics and tissue engineering blocks of the course.

Questions:

1. **Prior to conducting any additional reading or preparation for this problem set and without consulting any references**, brainstorm the challenges of developing a synthetic cartilage replacement. Consider the nature of the load in terms of magnitude, time, and direction; reasonable stress and strain magnitudes; energy dissipation; candidate materials and/or material strategies to accommodate stresses and strains; and simplifying assumptions to aid in modeling. Your answer should be no more than 2 pages long.
2. Proteoglycans are proteins in the porous extracellular matrix of connective tissues. Using the graphic below for the charge density of proteoglycans and collagen across a pH range, (a) sketch the net charge of the tissue versus bath pH over the entire pH 2-10 range; (b) sketch the dependence of the equilibrium elastic modulus E on pH for the pH 2-10 range; and (c) repeat (b) for the case in which proteolytic enzymes (e.g. caused by disease or purposely in animal models of disease) have removed the proteoglycan constituents, leaving the collagen network intact.



3. Derive the time dependent Maxwell and Kelvin-Voight viscoelastic expressions for creep and stress relaxation. Graphically compare the differences between Maxwell, Kelvin-Voight, and the Standard Linear Solid (Maxwell form) models for time dependent creep and stress relaxation using a range of spring and damping constants that achieve the mechanical response for the loads, strain displacements and time response identified in Problem 1, i.e. for synthetic cartilage replacement. Explain the difference in the time dependent creep and stress-relaxation responses between the three models.

Guidance:

1. Clearly state all assumptions for all answers.
2. Include derivations for all relationships that are modeled.
3. Clearly label and annotate figures, tables, and schematics.
4. Derivations and schematics may be hand written as long as they are neatly presented.

Report Format:

1. References. Use the CSE citation-sequence. See the following site as a guide, <http://www.wsulibs.wsu.edu/quickguides/cse>.

Instructor Solution

NOT PROVIDED



PROBLEM SET #3

Handouts

PROBLEM SET 3: Tissue Engineering**Tissue Growth Model****References:**

The following references are intended to serve as a starting point to analyze the questions below. You may use any additional references and you are encouraged to collaborate with your classmates. Ensure that all referenced material and collaboration is properly documented.

1. Introduction to Biomedical Engineering, Chapter 6. Tissue Engineering
2. A 3D Hybrid Model for Tissue Growth: The Interplay between Cell Population and Mass Transport Dynamics, Cheng (2009).
3. Adipose-Derived Stem Cells in Functional Bone Tissue, Bodle (2011).
4. Armed Forces Institute of Regenerative Medicine Annual Report (2012).
5. Bioreactors for tissue mass culture: Design, characterization, and recent advances. Martin (2005).

All the supplementary literatures are placed in the problem set submission portal.

Background:

The past nearly two decades of sustained combat operations has resulted in veterans returning home with various injuries. Severe burns, missing limbs, organ damage, and major craniofacial damage are just some of the areas in need of major medical advancement. It is a complex area to address since multiple levels and tissue types are involved and require different strategies for repair. Whether for an approximate 2-dimensional “sheet” of tissue, or a complex, 3-dimensional organ, each tissue type comes with its own set of challenges, ranging from cell seeding to biocompatibility.

This problem set is intended to approach these challenges by applying basic cell growth and mass transport concepts to a 3-dimensional tissue.

Questions:

1. Describe the considerations and attendant challenges in growing a 3-dimensional engineered tissues. This should be no longer than two pages.
2. Develop a model to determine the time it takes from seeding a 3-dimensional scaffold to achieving complete cell density. Using a single cell type, model output should include the following plots: (a) cell growth versus time; (b) growth time to achieve monolithic tissue as a function of initial number of cells, and scaffold pore size. Using stem cell differentiation, repeat the model growth curves (a) and (b) above starting with pluripotent cells and growing through until all cells are differentiated.
3. For a 2 mm cube volume element, use Ficks Second Law in one dimension to model the mass transport of glucose from the surface to the center of the volume. Model output should depict glucose concentration as a function of distance and time – show with both a 2D and 3D plot. How do these

plots change as the bulk concentration of glucose (boundary condition) increases and decreases? How do they change as the volume element scaffold density changes (i.e. diffusivity)?

Guidance:

1. Clearly state all assumptions for all answers.
2. Include derivations for all relationships that are modeled.
3. Clearly label and annotate figures, tables, and schematics.
4. Derivations and schematics may be hand written as long as they are neatly presented.

Report Format:

1. References. Use the CSE citation-sequence. See the following site as a guide, <http://www.wsulibs.wsu.edu/quickguides/cse>.

Instructor Solution

NOT PROVIDED



PROBLEM SET #4

Handouts

PROBLEM SET 4: Compartment Modeling & Enzyme Kinetics**References:**

The following references are intended to serve as a starting point to analyze the questions below. You may use any additional references and you are encouraged to collaborate with your classmates. Ensure that all referenced material and collaboration is properly documented.

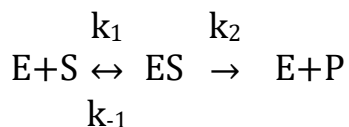
1. Introduction to Biomedical Engineering, Chapter 7; Chapter 8.
2. Biochemistry, 7th ed., Berg, Chapter 8 (or similar Biochemistry text for enzyme kinetics).
3. Introduction to Clinical Pharmacokinetics, Thompson (2000).

Background:

To determine the appropriate dose of medicine to administer, many factors come into play: the required drug concentration at the site of interest (minimum effective concentration – MEC); the kinetics and biochemical mechanism of the drug; and the mass transport of the drug to the site of interest. In this problem set you will attempt to determine the minimum effective concentration of a competitive inhibitory drug. The MEC will be used in a mass transport model to determine the required drug concentration in blood plasma. The required drug concentration in the blood plasma will be used in a one-compartment model to determine the drug delivery dose and frequency.

Questions:

1. Derive the Michaelis-Menton equation for the reaction below. The final form of the equation should be V_o as a function of the terms V_{max} , $[S]$ and K_M .



2. Derive the Michaelis-Menton for the reaction above in the presence of a competitive inhibitor. The final form of the equation should be V_o as a function of the terms V_{max} , $[S]$, $[I]$, K_M and K_i . Plot the dependence of V_o on $[I]$ as a function of a range of $[S]$ (use a double reciprocal plot).
3. A competitive inhibitor (I) is administered with the intent of decreasing the rate of the reaction catalyzed by E by 70% as a means of decreasing tumor growth. For nominal fixed values of $[S] = 80$ mg/ml and $[E] = 5$ ng/ml, determine the drug concentration, $[I]$, required at the tumor site.
4. Using a one-compartment model for blood plasma, model a drug dosage regimen that achieves the steady-state concentration determined in Problem 3.
5. What are factors that the models do not consider that would impact the actual dosing regimen?

Guidance:

1. Clearly state all assumptions for all answers.
2. Include derivations for all relationships that are modeled.
3. Clearly label and annotate figures, tables, and schematics.
4. Derivations and schematics may be hand written as long as they are neatly presented.

Report Format:

1. References. Use the CSE citation-sequence. See the following site as a guide,
<http://www.wsulibs.wsu.edu/quickguides/cse>.

Instructor Solution

NOT PROVIDED



PROBLEM SET #5

Handouts

PROBLEM SET 5: Biochemical Engineering**References:**

The following references are intended to serve as a starting point to analyze the questions below. You may use any additional references and you are encouraged to collaborate with your classmates. Ensure that all referenced material and collaboration is properly documented.

1. Construction of a genetic toggle switch in *Escherichia coli*, Gardner (2000).
2. Programming and engineering biological networks, Chin (2006)
3. Any differential calculus textbook describing slope fields.

Background:

With the desire to develop synthetic circuits within existing biological systems and ultimately artificial cellular systems, each circuit component must be developed using biological elements with the ability to tune the inputs and outputs within an appropriate timeframe. Garner, Cantor, and Collins developed a toggle switch in *E.coli* that allows for an inducible binary output. The objective of this problem set is to understand the model construct, mathematically model the output, and evaluate the utility of the circuit for actual implementation.

Questions:

1. Explain the toggle switch construct used including the concept, math model and variables, actual biological elements used, the inducers, and the output.
2. Using modeling software, reproduce the results for Figures 2a, 2b, and 2d. Your graphical output should look identical to the figures in the paper and directly plot from your math model. Include an additional plot that depicts the slope field for 2a and explain its significance to the intent of a toggle switch.
3. Based on the experimental data presented in the paper, evaluate the utility of such a toggle switch in an actual synthetic system and as part of a more complex synthetic circuit.

Guidance:

1. Clearly state all assumptions for all answers.
2. Include derivations for all relationships that are modeled.
3. Clearly label and annotate figures, tables, and schematics.
4. Derivations and schematics may be hand written as long as they are neatly presented.

Report Format:

1. References. Use the CSE citation-sequence. See the following site as a guide, <http://www.wsulibs.wsu.edu/quickguides/cse>.

Instructor Solution

NOT PROVIDED



TAB J Capstone PROJECT

Handouts

Capstone Paper (150 points)
IPR Presentation (50 Points)
Capstone Presentation (50 points)

IPR Presentation Due NLT 0740 26 MARCH 26

Paper Due NLT 2359 8 MAY 26

Presentation Due NLT 0740 5 MAY 26

Purpose:

The purpose of the design project and presentation is to allow you to further explore and develop a bioengineering model of the system of your choice. You may expand upon a model you developed in any of the previous problem sets, or develop a bioengineering model of a system that interests you and one that may support a research course project.

If you use a previously developed model from a problem set, parameter variation may be included, but does not meet the intent of further development, which should entail adding complexity to account for previously disregarded or simplified factors.

Format:

The **written project submission** should include:

Background

Initial Model (if used from a previous problem set), or established models from the work of other researchers.

Model Development – this is the main effort of the project

Discussion

Annexes – computer code and derivations as appropriate

IPR Presentation

Plan for no 5 minutes of presentation.

No more than 3 Power Point slides, excluding title slide.

Slides should showcase your model topic, concept, and initial results.

Final Presentation

Plan for no 7 minutes of presentation and 3 minutes of questions.

No more than 5 Power Point slides, excluding title slide.

Slides should showcase your model development and results.

Final presentations will be given in-person on Lesson 29 and 30 (5 MAY 26 and 7 MAY 26).

Guidance:

1. Clearly state all assumptions for all answers.
2. Include derivations for all relationships that are modeled.
3. Clearly label and annotate figures, tables, and schematics.
4. Derivations and schematics may be hand written as long as they are neatly presented.

Report Format:

1. References. Use the CSE citation-sequence. See the following site as a guide,
<http://www.wsulibs.wsu.edu/quickguides/cse>.

Instructor's Name: Dr. Simuck F. Yuk				Cadet Assessed: CDTs				
Course: CH450, AY26-2				Major of Person Assessed: Chemical Engineering (CEN1)				
	1 – Needs Improvement	1	2	3 – Meets Expectations	3	4	5 – Exceeds Expectations	5
Technical Competence	Some misunderstandings of the technical content.			Demonstrated knowledge of the technical content adequate to accomplish task.			Exceptional knowledge of technical content.	
Communication	Did not provide specific suggestions for improvement to others. Unable or unwilling to share information.			Effectively communicated important points to others. Facilitated sharing of information.			Exceptional ability to explain important points. Very effectively communicated ideas for improvement. Shared information effectively.	
Organization	Was not prepared or did not give sufficient time to prepare. Unaware of suspenses or progress.			Demonstrated organization and awareness of suspenses and progress.			Exceptional organization of the team and assignments of tasks to accomplish the mission in the most efficient manner possible.	
	Comments:							
	Final Score: /100							